



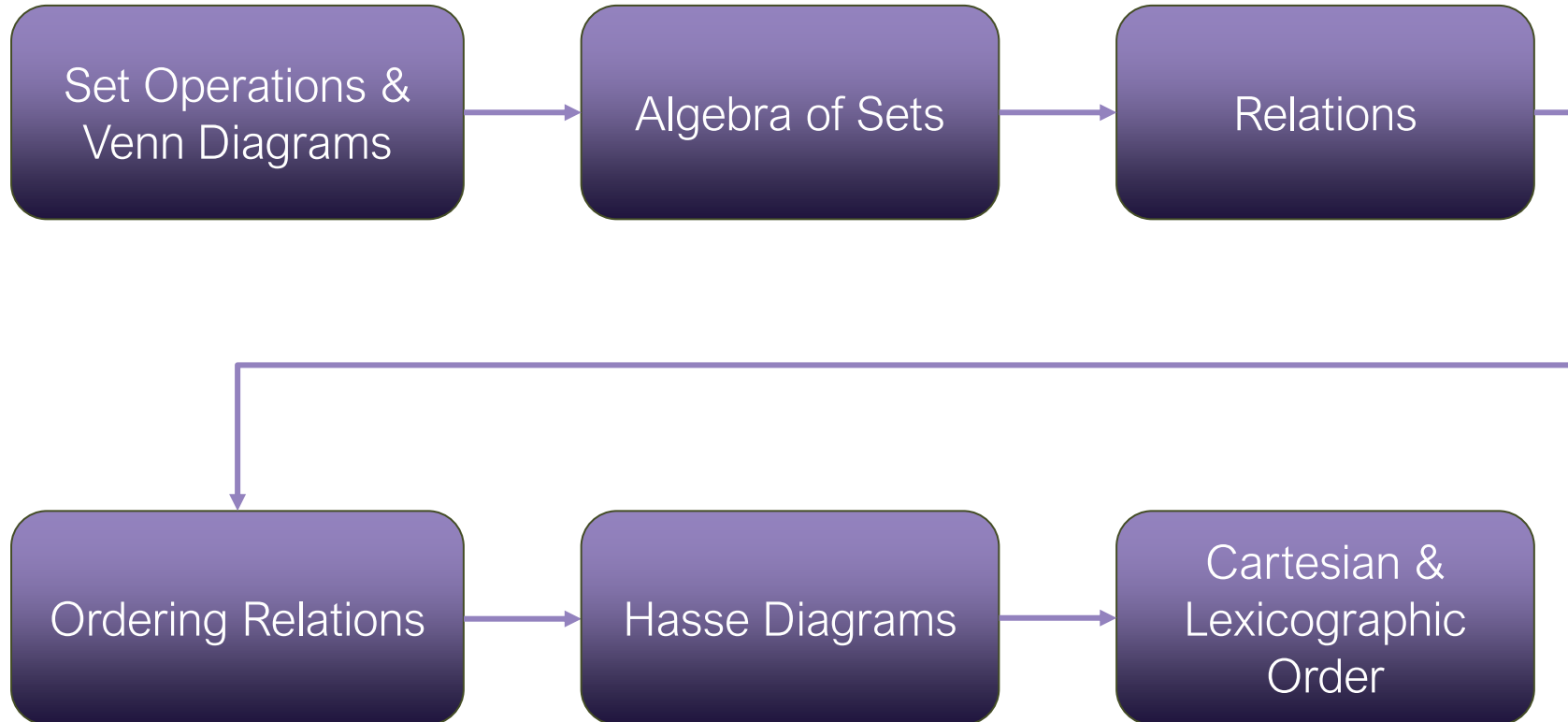
Logic & Sets Day 2

Midterm course | Mara Spadon | 2023

Planning

Sunday 	Logic week 1 + 2 + 3
Tuesday	Sets week 1: ○ Set operations ○ Venn diagrams Sets week 2: ○ Algebra of sets ○ Relations Sets week 3: ○ Ordering relations ○ Hasse Diagrams ○ Cartesian and Lexicographic Order

Planning



Set Operations & Venn Diagrams

Definition:

A set is an unordered collection of elements

Examples:

- DaysOfWeek = {Mon, Tue, Wed, Thu, Fri, Sat, Sun}
- A = {1, 4, 2, 6}

Often used: $\rightarrow$ pos

- $\mathbb{N} = \{0, 1, 2, \dots\}$ (Natural numbers)
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ (Integer numbers)

$\leftarrow$ neg $\rightarrow$ pos

$\mathbb{R}$ = set of all Real number $= \sqrt{3}, 8.5,$

Definition:

A set is an unordered collection of elements

Examples:

- DaysOfWeek = {Mon, Tue, Wed, Thu, Fri, Sat, Sun}
- A = {1, 4, 2, 6}

Often used:

- $\mathbb{N} = \{0, 1, 2, \dots\}$ (Natural numbers)
- $\mathbb{Z} = \{\dots, -2, 1, 0, 1, 2, \dots\}$ (Integer numbers)

MultiplesOf2 := $\{2k : k \text{ a natural number}\} = \{0, 2, 4, \dots\}$

Months := $\{x : x \text{ is a month}\}$

↪ where

Definition:

A set is an unordered collection of elements

Element

$a \in A$	means	a is an element of the set A
$a \notin A$	means	a is <u>not</u> an element of the set A

Subset

$A \subseteq B$	means	all elements of A are elements of B
$A \not\subseteq B$	means	<u>at least one</u> element of A is <u>not</u> in B

Examples

$4 \in \{1, 2, 3, \underline{4}\}$	$\{2, 3\} \subseteq \{1, \underline{2}, 3, 4\}$
$5 \notin \{1, 2, 3, 4\}$	$\{\underline{2}, 5\} \not\subseteq \{1, 2, 3, 4\}$

Definition:

A set is an unordered collection of elements

Set equality:

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$

$$A = \{1, 2\}, B = \{1, 2\}$$

$$A \subseteq B$$

$$B \subseteq A$$

$$\hookrightarrow A = B$$

Definition:

A set is an unordered collection of elements

Set equality:

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$

Empty set:

$\emptyset$ or $\{ \}$ has no elements



Is a subset of every set!

$$\emptyset \subseteq A$$

$$\emptyset \subseteq B$$

Definition:

A set is an unordered collection of elements

Set equality:

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$

Empty set:

$\emptyset$ or $\{ \}$ has no elements $\longrightarrow$ Is a subset of every set!

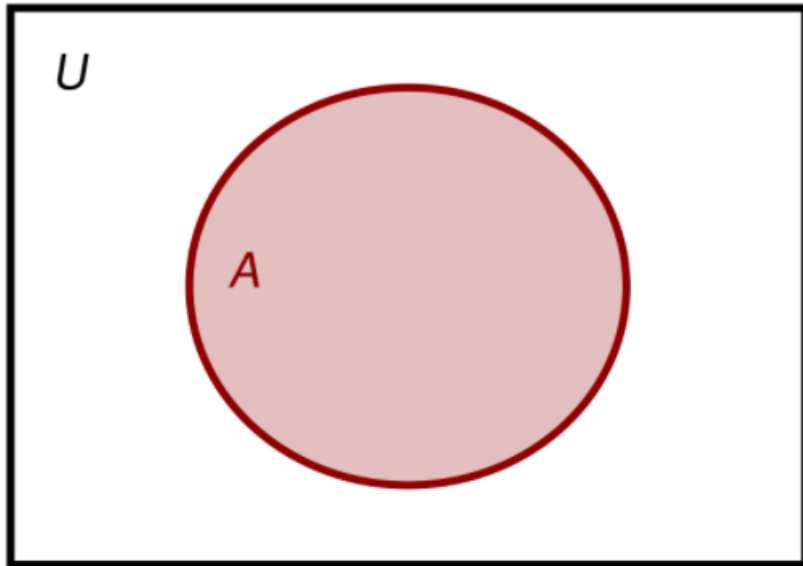
Number of elements:

$\#\{4, 3, 3, 1, 3, 2\} = \#\{1, 2, 3, 4\} = 4$

$\hookrightarrow$ no duplicates

Sets

Visualization:



All sets are a subset of the Universe U

$$A \subseteq U$$

Set Operations and Venn Diagram

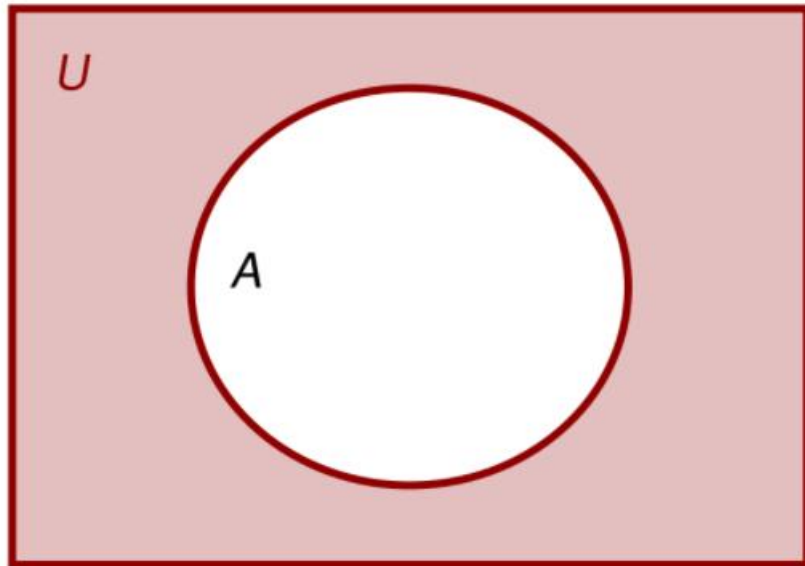
Operations on sets:

- Complement → Similar to negation
- Union → 'or'
- Intersection → 'and'
- Minus
- Symmetric Difference → 'exclusive or'

Set Operations and Venn Diagram

Visualization using Venn Diagrams!

Complement of A



Example

U : all 4-letter words

A : words with a

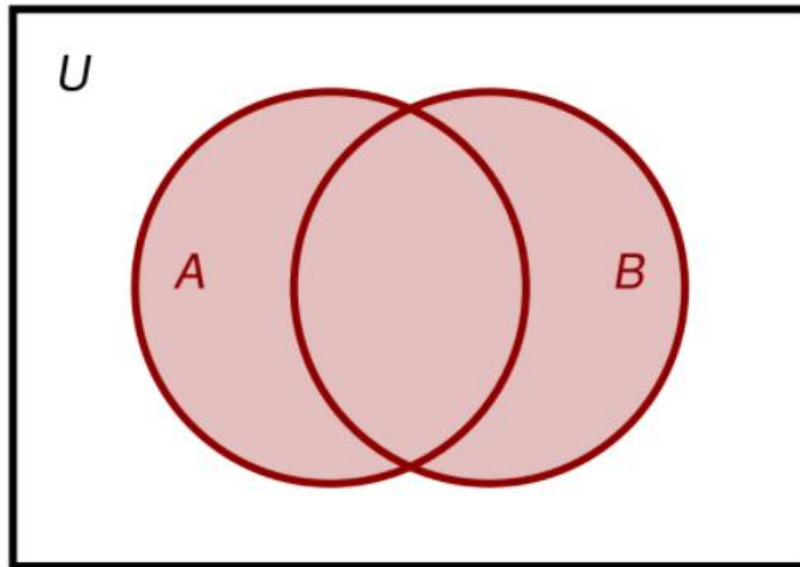
A' :

words with *no* a

$$A' := \{x \in U : x \notin A\}$$

Set Operations and Venn Diagram

Union of A and B



Example

U : all 4-letter words

A : words with a

B : words with b

$A \cup B$:

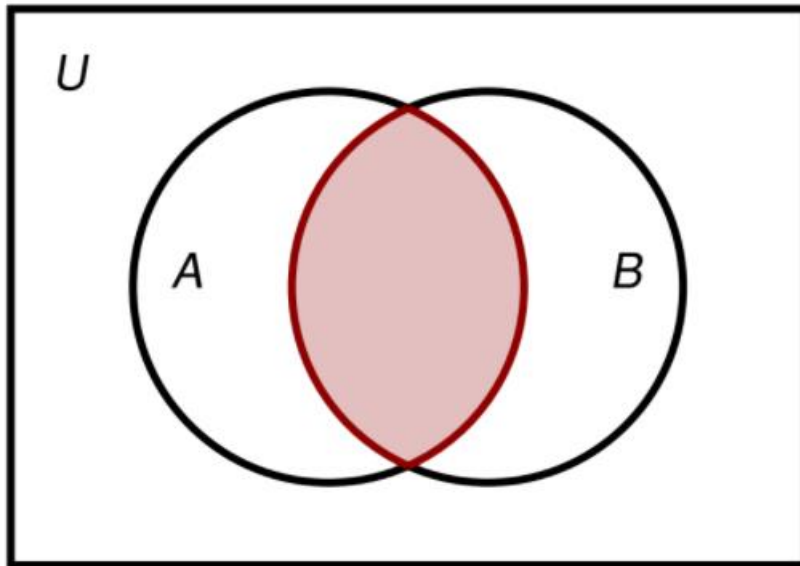
words with a or b

(inclusive or)

$$A \cup B := \{x \in U: \underline{x \in A} \text{ or } \underline{x \in B}\}$$

Set Operations and Venn Diagram

Intersection of A and B



Example

U : all 4-letter words

A : words with a

B : words with b

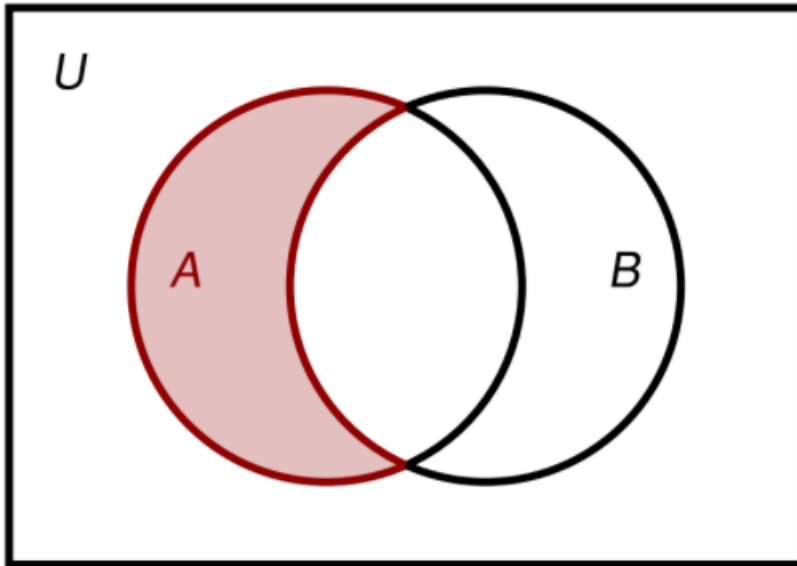
$A \cap B$:

words with a and b

$$A \cap B := \{x \in U: \underline{x \in A} \text{ and } \underline{x \in B}\}$$

Set Operations and Venn Diagram

A minus B



Example

U : all 4-letter words

A : words with a

B : words with b

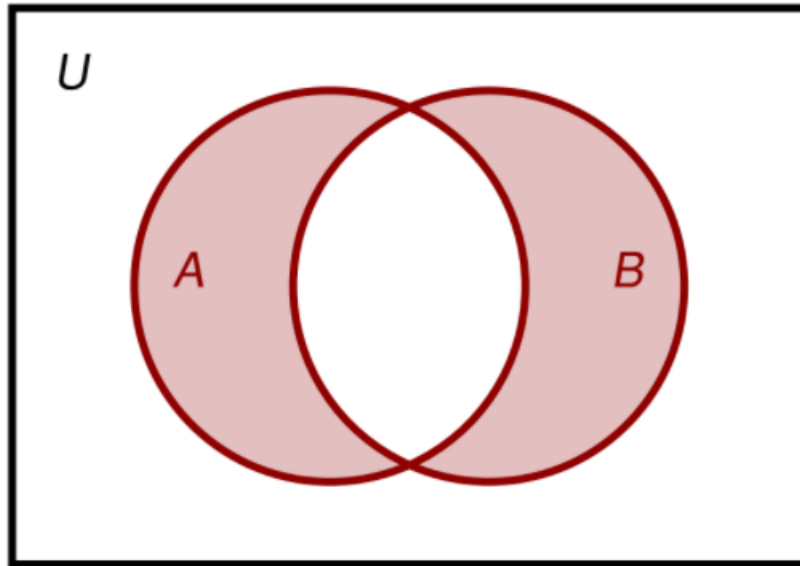
$A \setminus B$:

words with a *but no* b

$$A \setminus B := \{x \in A : x \notin B\} = A \cap B'$$

Set Operations and Venn Diagram

Symmetric difference of A and B



Example

U : all 4-letter words

A : words with a

B : words with b

$A \triangle B$:

words with a *or* b

but not both

(exclusive or)

$$\begin{aligned} \underline{A \triangle B} &:= (A \cup B) \cap (A \cap B)' \\ &= (A \setminus B) \cup (B \setminus A) \end{aligned}$$

Set Operations and Venn Diagram: Example

Show that $A \cap (B \setminus C) = (A \cap B) \setminus C$, using a sequence of Venn Diagrams

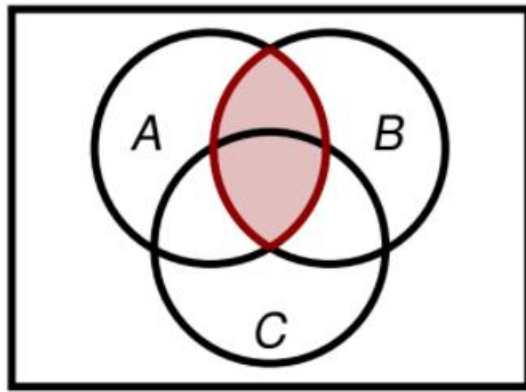
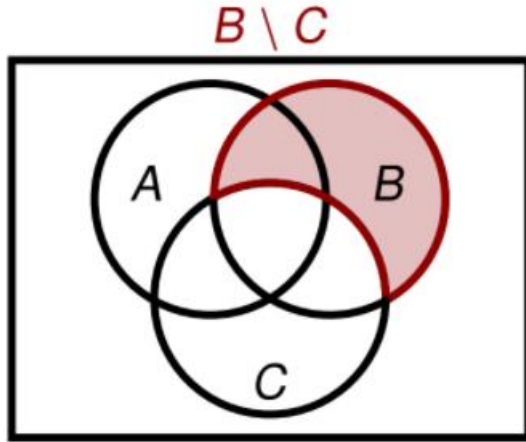
Set Operations and Venn Diagram: Example

Show that $A \cap (B \setminus C) = (A \cap B) \setminus C$, using a sequence of Venn Diagrams

1. For each side of the equality:
Draw a Venn diagram for the simplest part of the formula
2. Draw a new Venn diagram that extends the previous diagram with the next part of the formula.
Repeat until you have Venn diagrams of both sides of the formula!
3. Then compare them!

Set Operations and Venn Diagram: Example

Show that $A \cap (B \setminus C) = (A \cap B) \setminus C$, using a sequence of Venn Diagrams

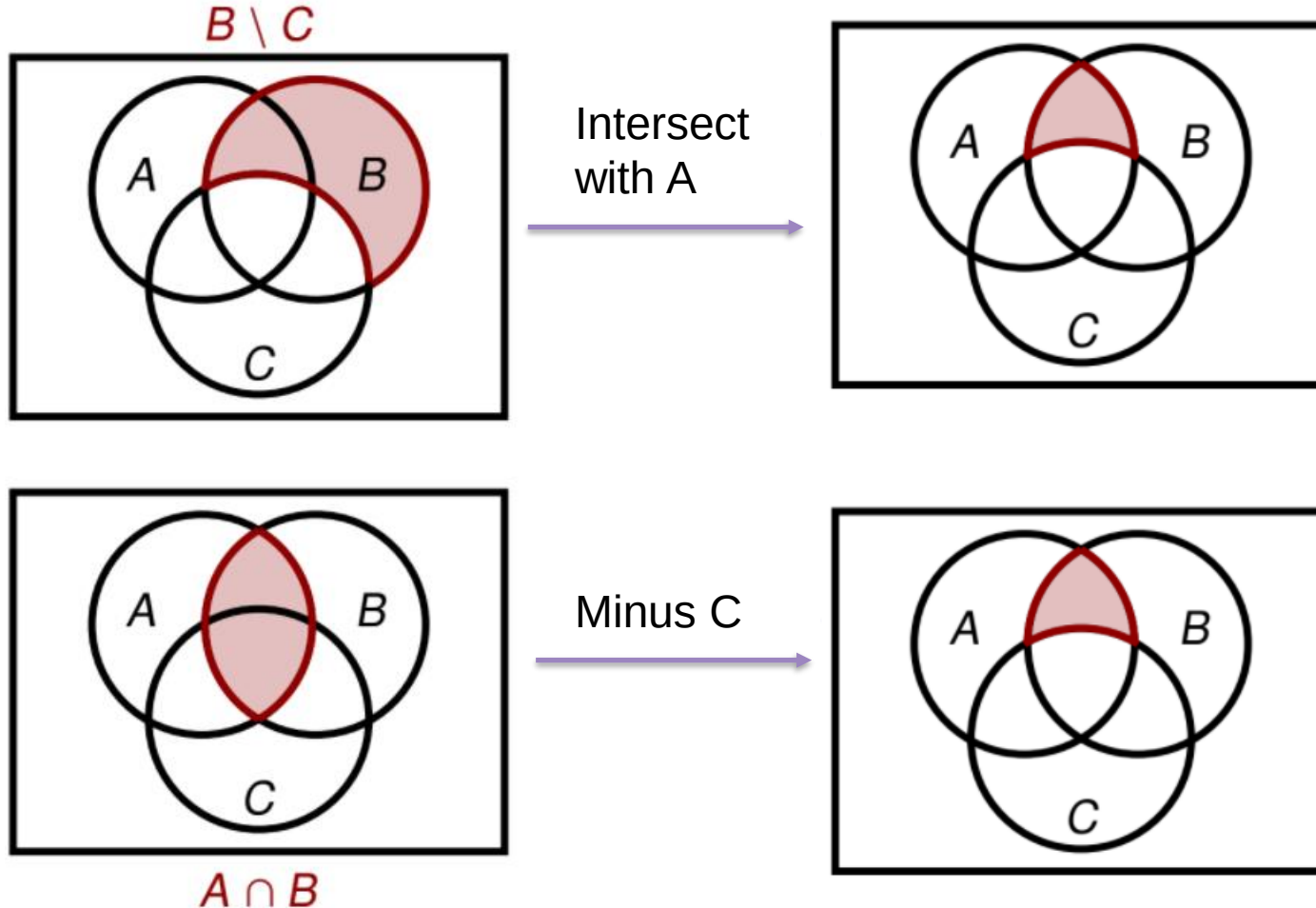


$A \cap B$

1. For each side of the equality:
Draw a Venn diagram for the simplest part of the formula
2. Draw a new Venn diagram that extends the previous diagram with the next part of the formula. Repeat until you have Venn diagrams of both sides of the formula!
3. Then compare them!

Set Operations and Venn Diagram: Example

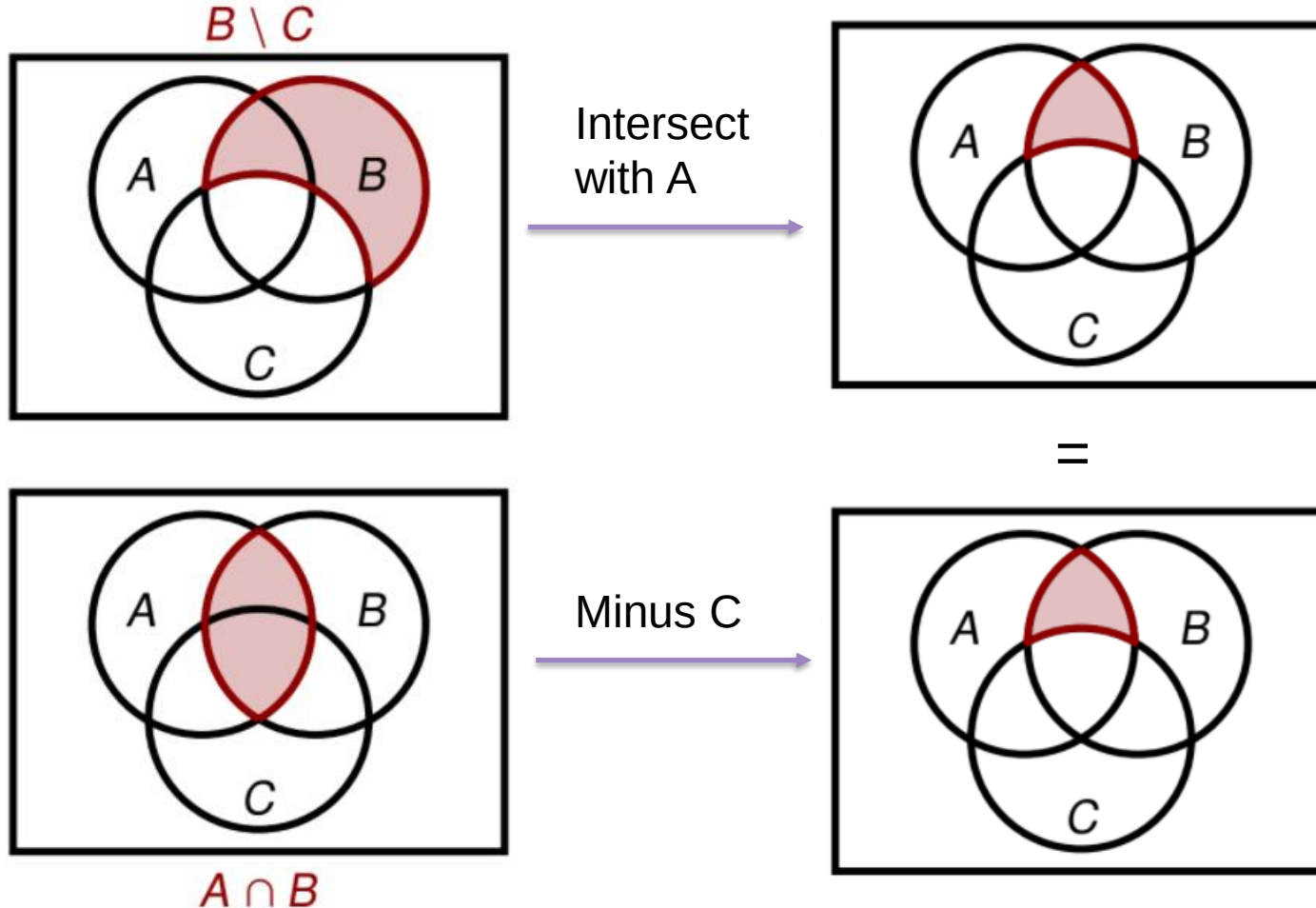
Show that $A \cap (B \setminus C) = (A \cap B) \setminus C$, using a sequence of Venn Diagrams



1. For each side of the equality:
Draw a Venn diagram for the simplest part of the formula
2. **Draw a new Venn diagram that extends the previous diagram with the next part of the formula. Repeat until you have Venn diagrams of both sides of the formula!**
3. Then compare them!

Set Operations and Venn Diagram: Example

Show that $A \cap (B \setminus C) = (A \cap B) \setminus C$, using a sequence of Venn Diagrams



1. For each side of the equality:
Draw a Venn diagram for the simplest part of the formula
2. Draw a new Venn diagram that extends the previous diagram with the next part of the formula. Repeat until you have Venn diagrams of both sides of the formula!
3. **Then compare them!**

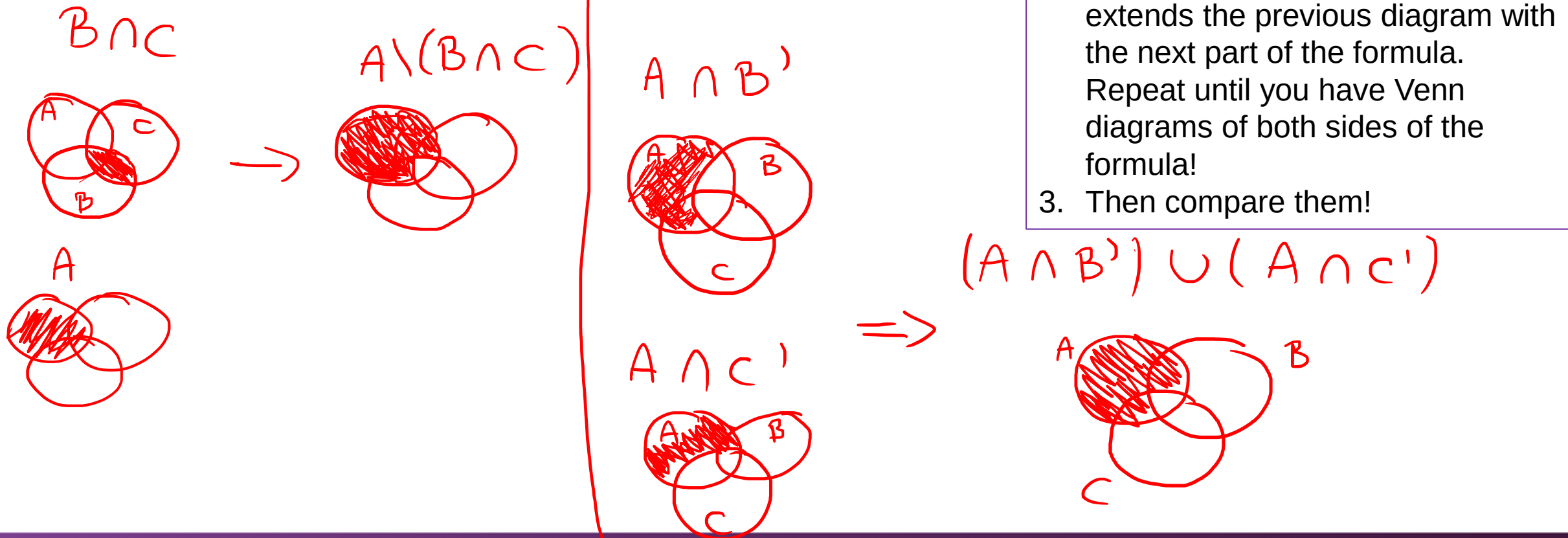
Set Operations and Venn Diagram: Exercise!

Exam Question: midterm 2021

5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$\underline{A \setminus (B \cap C) = (A \cap B') \cup (A \cap C')}.$$

a) Prove this identity using a sequence of Venn diagrams.



1. For each side of the equality: Draw a Venn diagram for the simplest part of the formula
2. Draw a new Venn diagram that extends the previous diagram with the next part of the formula. Repeat until you have Venn diagrams of both sides of the formula!
3. Then compare them!

Set Operations and Venn Diagram: Exercise!

Exam Question: midterm 2021

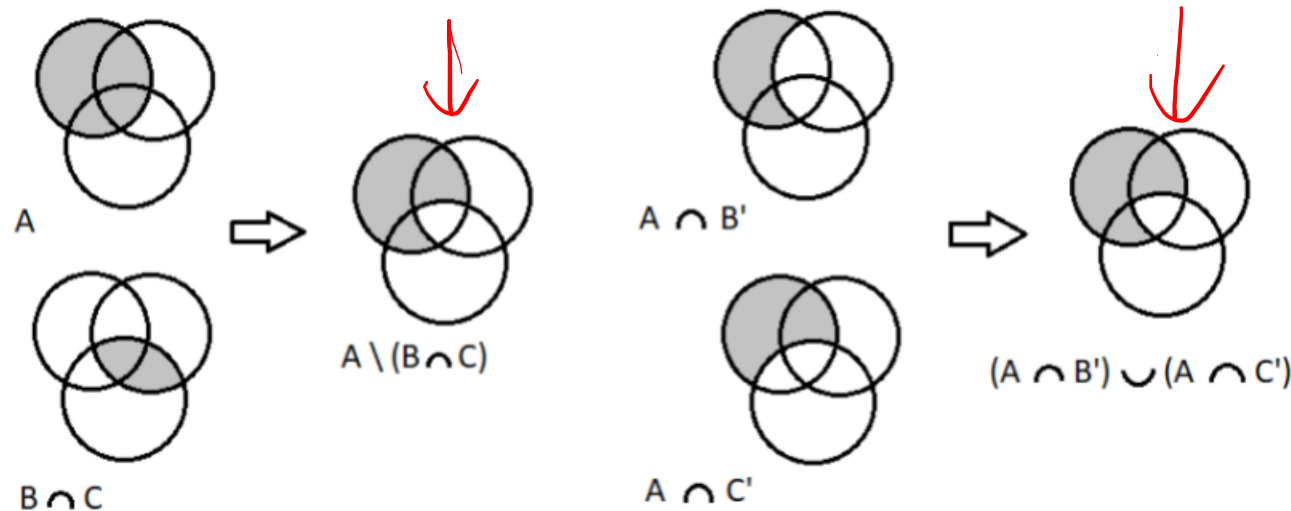
5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \setminus (B \cap C) = (A \cap B') \cup (A \cap C').$$

a) Prove this identity using a sequence of Venn diagrams.

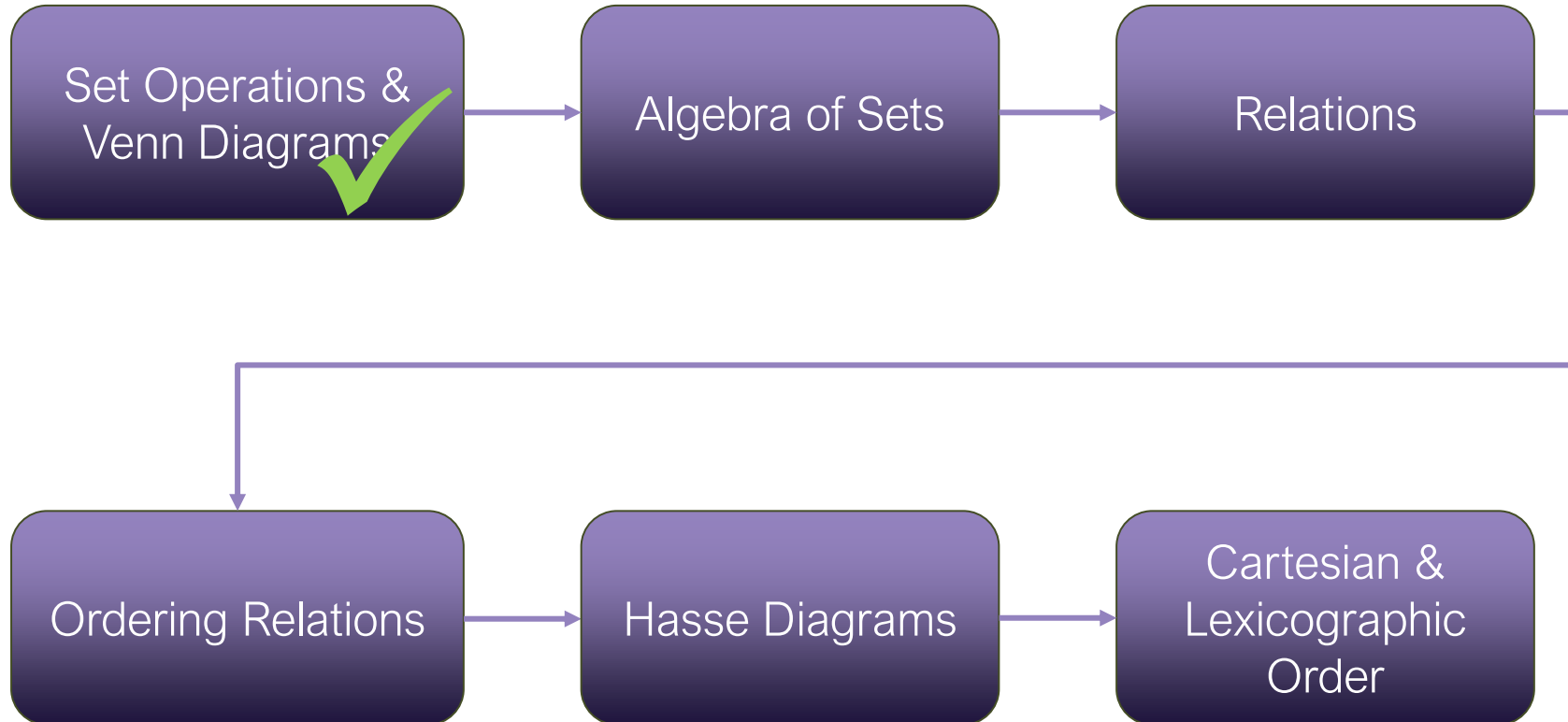
Answer:

a) Using Venn diagrams we see that



1. For each side of the equality:
Draw a Venn diagram for the simplest part of the formula
2. Draw a new Venn diagram that extends the previous diagram with the next part of the formula.
Repeat until you have Venn diagrams of both sides of the formula!
3. Then compare them!

Planning



Algebra of Sets

Algebra of sets

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Complement

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

Idempotence

$$A \cap A = A$$

$$A \cup A = A$$

De Morgan

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Associativity

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

$$A \setminus B = A \cap B'$$

$$A \triangle B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Example

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

The algebra of sets

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Associativity

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Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

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Involution

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Definitions

$$A \setminus B = A \cap B'$$

$$A \Delta B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Example

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$(A' \cap B)' \cup (C \cap A')'$$

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Algebra of sets: Example

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$\begin{aligned} & (A' \cap B)' \cup (C \cap A')' \\ &= ((A')' \cup B') \cup (C' \cup (A')') \quad (\text{De Morgan}) \end{aligned}$$

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

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Involution

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Definitions

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$$A \Delta B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Example

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$\begin{aligned}
 &(A' \cap B)' \cup (C \cap A')' \\
 &= ((A')' \cup B') \cup (C' \cup (A')') \quad (\text{De Morgan}) \\
 &= (A \cup B') \cup (C' \cup A) \quad (\text{involution})
 \end{aligned}$$

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Distributivity

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$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Example

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$\begin{aligned}
 &(A' \cap B)' \cup (C \cap A')' \\
 &= ((A')' \cup B') \cup (C' \cup (A')') && \text{(De Morgan)} \\
 &= (A \cup B') \cup (C' \cup A) && \text{(involution)} \\
 &= (A \cup A) \cup (B' \cup C') && \text{(reorder)}
 \end{aligned}$$

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Identity

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Domination

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Involution

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Definitions

$$A \setminus B = A \cap B'$$

$$A \Delta B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Example

Example

Check that $(A' \cap B)' \cup (C \cap A')' = \underline{A \cup (B \cap C)'}$

Solution

$$\begin{aligned}
 & (A' \cap B)' \cup (C \cap A')' \\
 &= ((A')' \cup B') \cup (C' \cup (A')') && \text{(De Morgan)} \\
 &= (A \cup B') \cup (C' \cup A) && \text{(involution)} \\
 &= (A \cup A) \cup (B' \cup C') && \text{(reorder)} \\
 &= A \cup (B' \cup C') && \text{(idempotence)}
 \end{aligned}$$

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$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

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Algebra of sets: Example

The algebra of sets

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$\begin{aligned}
 &(A' \cap B)' \cup (C \cap A')' \\
 &= ((A')' \cup B') \cup (C' \cup (A')') \\
 &= (A \cup B') \cup (C' \cup A) \\
 &= (A \cup A) \cup (B' \cup C') \\
 &= A \cup (B' \cup C') \\
 &= A \cup (B \cap C)'
 \end{aligned}$$

$$= A \cup (B \cap C)'$$

write down
names
↓

(De Morgan)

(involution)

(reorder)

(idempotence)

(De Morgan)

Commutativity

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$$(A \cap B)' = A' \cup B'$$

$$A' \cup B' = (A \cap B)'$$

Algebra of sets: Exercise!

Exam Question: midterm 2021

5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \setminus (B \cap C) = (A \cap B') \cup (A \cap C').$$

~~a) Prove this identity using a sequence of Venn diagrams.~~

b) Prove it using the rules of the algebra of sets.

$$\begin{aligned}
 A \setminus (B \cap C) &= A \cap (B \cap C)' && \text{Definitions} \\
 &= A \cap (B' \cup C') && \text{De Morgan} \\
 &= (A \cap B') \cup (A \cap C') && \text{Distributivity}
 \end{aligned}$$

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

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$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

$$A \setminus B = A \cap B'$$

$$A \triangle B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Exercise!

Exam Question: midterm 2021

5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \setminus (B \cap C) = (A \cap B') \cup (A \cap C').$$

- a) ~~Prove this identity using a sequence of Venn diagrams.~~
 b) Prove it using the rules of the algebra of sets.

Answer:

$$\begin{aligned}
 \text{b)} \quad A \setminus (B \cap C) &= A \cap (B \cap C)' && \text{(Definitions)} \\
 &= A \cap (B' \cup C') && \text{(De Morgan)} \\
 &= (A \cap B') \cup (A \cap C') && \text{(Distributivity)}
 \end{aligned}$$

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

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Complement

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Definitions

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$$A \triangle B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Algebra of sets: Exercise 2!

Exam Question: resit 2021

5. Sets (6+6 points)

Consider the following set-theoretic equality

$$(((A' \cap C) \cup C') \cap B)' = B' \cup (A \cap C).$$

(a) Prove it using the rules of the algebra of sets.

$$\begin{aligned}
 & (((A' \cap C) \cup C') \cap B)' \\
 &= ((A' \cap C) \cup C')' \cup B' \quad \text{De Morgan} \\
 &= ((A' \cap C)' \cap C'') \cup B' \quad \text{De Morgan} \\
 &= ((A' \cap C)' \cap C) \cup B' \quad \text{involution} \\
 &= (A \cup C') \cap C \cup B' \quad \text{De Morgan} \\
 & \quad \text{involution}
 \end{aligned}$$

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Complement

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

Idempotence

$$A \cap A = A$$

$$A \cup A = A$$

De Morgan

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Associativity

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

$$A \setminus B = A \cap B'$$

$$A \Delta B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

DISTR:

$$((A \cap C) \cup (C' \cap C)) \cup B'$$

compl:

$$((A \cap C) \cup \emptyset) \cup B'$$

identity:

$$(A \cap C) \cup B'$$

reorder:

$$B' \cup (A \cap C)$$

Algebra of sets: Exercise 2!

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Complement

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

Idempotence

$$A \cap A = A$$

$$A \cup A = A$$

De Morgan

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Associativity

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

$$A \setminus B = A \cap B'$$

$$A \triangle B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

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Exam Question: resit 2021

5. Sets (6+6 points)

Consider the following set-theoretic equality

$$(((A' \cap C) \cup C') \cap B)' = B' \cup (A \cap C).$$

(a) Prove it using the rules of the algebra of sets.

Answer:

$$(((A' \cap C) \cup C') \cap B)' = \text{(De Morgan)}$$

$$((A' \cap C) \cup C')' \cup B' = \text{(De Morgan)}$$

$$((A' \cap C)' \cap C'') \cup B' = \text{(Involution)}$$

$$((A' \cap C)' \cap C) \cup B' = \text{(De Morgan)}$$

$$((A'' \cup C') \cap C) \cup B' = \text{(Involution)}$$

$$((A \cup C') \cap C) \cup B' = \text{(Distributivity)}$$

$$((A \cap C) \cup (C' \cap C)) \cup B' = \text{(Complement)}$$

$$((A \cap C) \cup \emptyset) \cup B' = \text{(Identity)}$$

$$(A \cap C) \cup B' = \text{(Reorder)}$$

$$B' \cup (A \cap C).$$

Algebra of sets: Exercise 2!

Exam Question: resit 2021

(6 points)

- (b) Assuming that $\#U = 14$, $\#A = 5$, $\#B = 7$, $\#C = 6$, $\#(A \cup C) = 9$, and $\#(A \cap B' \cap C) = 1$, compute the number of elements in $B' \cup (A \cap C)$.

Hint: it helps if you draw a Venn diagram!

$$\#B' \cup (A \cap C)$$

$$= \#B' + \#(A \cap C) - \#(A \cap B' \cap C)$$

$$= \#(U - B) + \#(A + C - (A \cup C)) - \#(A \cap B' \cap C)$$

$$= 14 - 7 + 5 + 6 - 9 - 1$$

$$= 8$$

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Complement

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

Idempotence

$$A \cap A = A$$

$$A \cup A = A$$

De Morgan

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Associativity

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

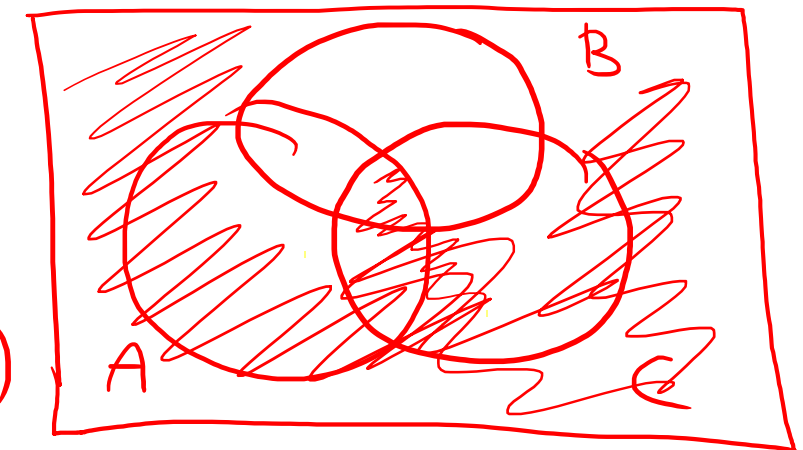
$$A \setminus B = A \cap B'$$

$$A \Delta B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$



Algebra of sets: Exercise 2!

The algebra of sets

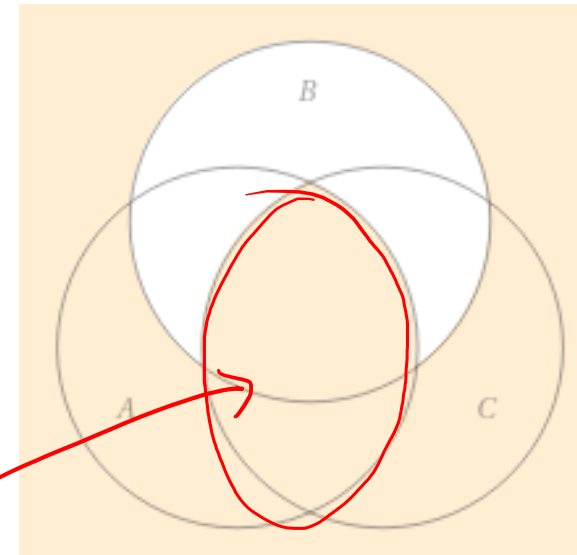
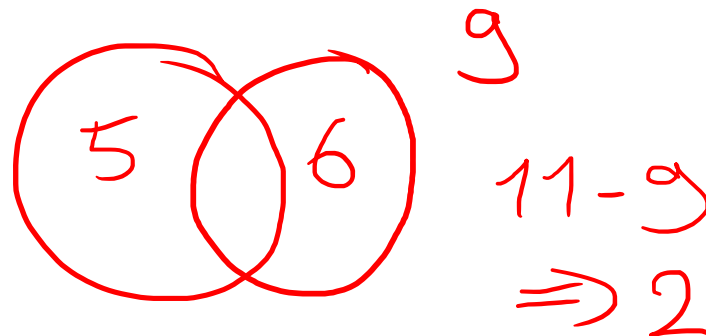
Exam Question: resit 2021

(b) Assuming that $\#U = 14$, $\#A = 5$, $\#B = 7$, $\#C = 6$, $\#(A \cup C) = 9$, and $\#(A \cap B' \cap C) = 1$, compute the number of elements in $B' \cup (A \cap C)$.

Hint: it helps if you draw a Venn diagram!

Answer:

$$\begin{aligned} \#(B' \cup (A \cap C)) &= \#B' + \#(A \cap C) - \#(A \cap B' \cap C) = \\ &= (\#U - \#B) + (\#A + \#C - \#(A \cup C)) - \#(A \cap B' \cap C) = \\ &= 14 - 7 + 5 + 6 - 9 - 1 = 8. \end{aligned}$$



Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Complement

$$A \cap A' = \emptyset$$

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Idempotence

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Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

$$A \setminus B = A \cap B'$$

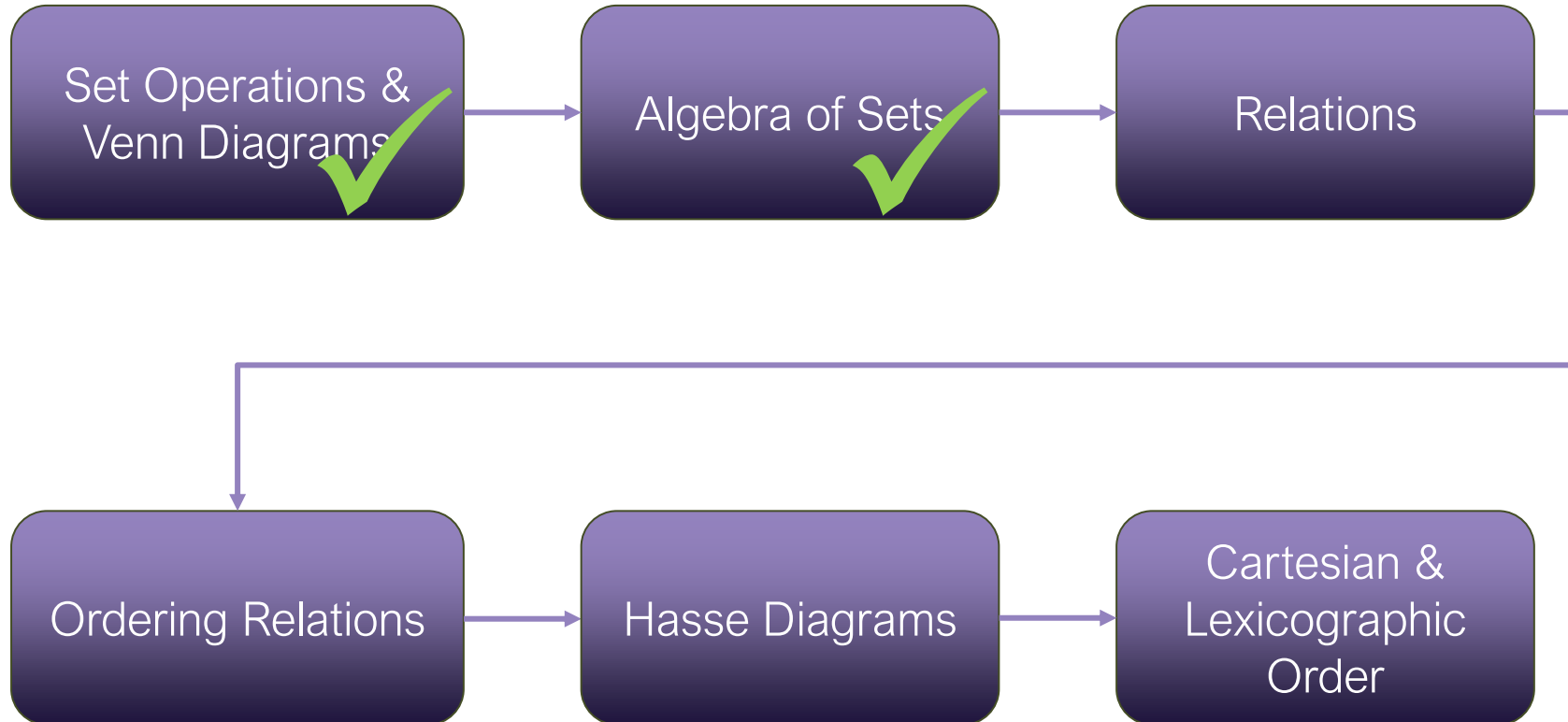
$$A \Delta B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Planning



Relations

Cartesian Product

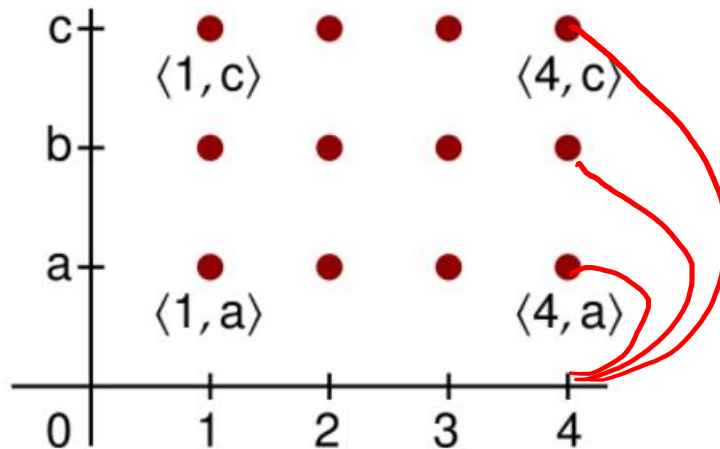
Cartesian product is the product of two (or multiple) sets

- $A \times B := \{ \langle a, b \rangle : a \in A \text{ and } b \in B \}$
- $A^2 := A \times A$

$\langle a, b \rangle$ is a set of pairs
(ordered list of length 2)

$$\{a, b, c\}^2 = \{ \langle a, a \rangle, \langle a, b \rangle, \langle a, c \rangle, \dots \}$$

Example



Cartesian product
 $\{1, 2, 3, 4\} \times \{a, b, c\}$

$$\{a, b, c\}$$

$$\{a, b, c\}$$

Relations

Relations show some relation between elements of sets

Examples

$IsBrotherOf := \{\langle x, y \rangle : x \text{ is brother of } y\} \subseteq People^2$

$FollowsCourse := \{\langle x, y \rangle : x \text{ follows } y\} \subseteq Students \times Courses$

Relations

Relations show some relation between elements of sets

Examples

$IsBrotherOf := \{\langle x, y \rangle : x \text{ is brother of } y\} \subseteq People^2$

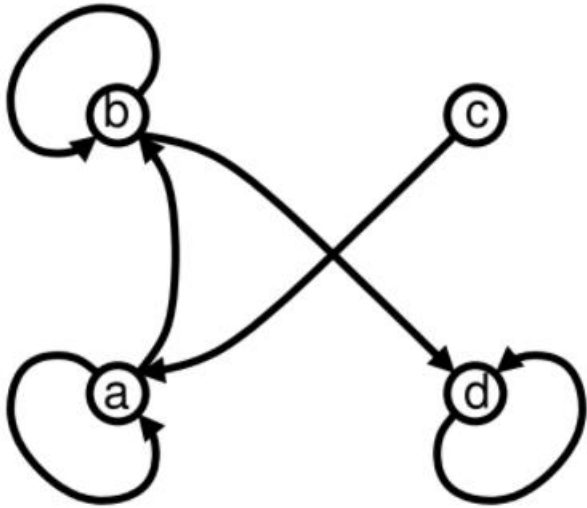
$FollowsCourse := \{\langle x, y \rangle : x \text{ follows } y\} \subseteq Students \times Courses$

Notation:

- We say $\langle x, y \rangle \in R$ if there is some relation R from x to y
- Alternatively: $x R y$

Relations

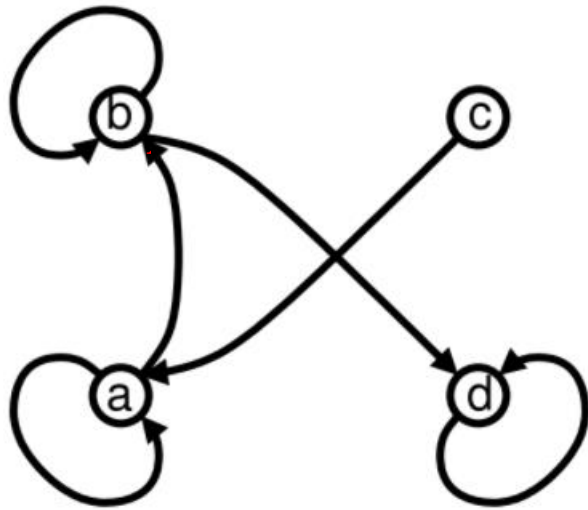
Directed graphs are used to visualize relations in a set A :



$$R = \{\langle a, a \rangle, \langle b, b \rangle, \langle d, d \rangle, \langle a, b \rangle, \langle b, d \rangle, \langle c, a \rangle\} \subseteq \{a, b, c, d\}^2$$

Relations

Directed graphs are used to visualize relations in a set A :



Matrix representation:

	a	b	c	d	(to)
a	1	1	0	0	
b	0	1	0	1	
c	1	0	0	0	
d	0	0	0	1	
(from)					

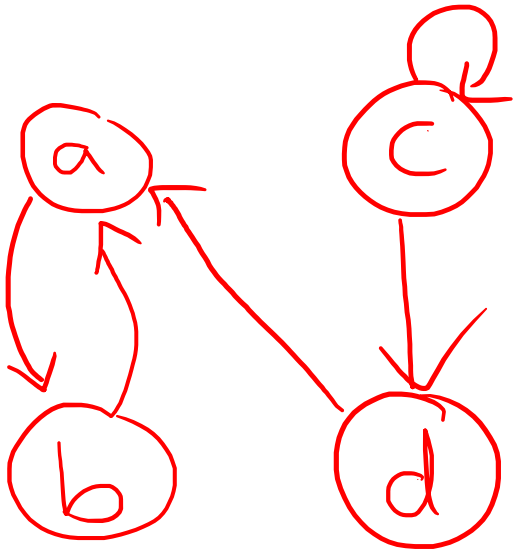
$$R = \{\langle a, a \rangle, \langle b, b \rangle, \langle d, d \rangle, \langle a, b \rangle, \langle b, d \rangle, \langle c, a \rangle\} \subseteq \{a, b, c, d\}^2$$

Relations: Exercise!

Draw the directed graph of the following relation R:

$$R = \{ \langle a, b \rangle, \langle b, a \rangle, \langle c, c \rangle, \langle c, d \rangle, \langle d, a \rangle \} \subseteq \{a, b, c, d\}^2$$

Write down the matrix representation as well



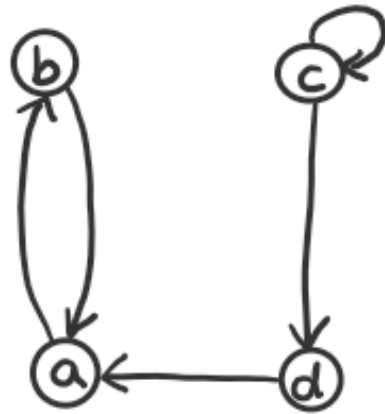
	a	b	c	d (to)
a	0	1	0	0
b	1	0	0	0
c	0	0	1	1
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(from)				

Relations: Exercise!

Draw the directed graph of the following relation R:

$$R = \{ \langle a, b \rangle, \langle b, a \rangle, \langle c, c \rangle, \langle c, d \rangle, \langle d, a \rangle \} \subseteq \{a, b, c, d\}^2$$

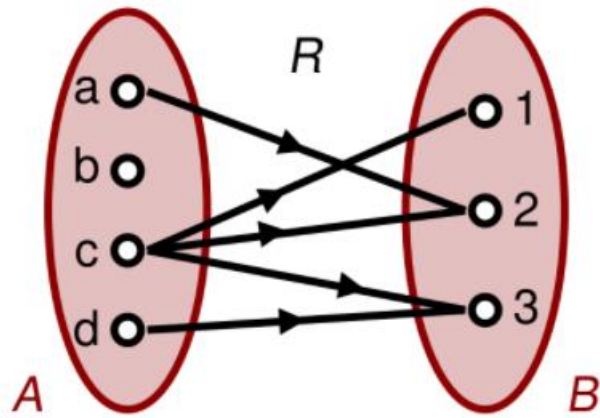
Write down the matrix representation as well



	a	b	c	d (to)
a	0	1	0	0
b	1	0	0	0
c	0	0	1	1
d	1	0	0	0
(from)				

Relations

Venn diagrams can be used to visualize all binary relations:



Matrix representation:

	1	2	3
a	0	1	0
b	0	0	0
c	1	1	1
d	0	0	1

$$R = \{\langle a, 2 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle, \langle d, 3 \rangle\} \subseteq \{a, b, c, d\} \times \{1, 2, 3\}$$

Inverse Relations

Definition:

The inverse of a relation R is $R^{-1} := \{ \langle x, y \rangle : \langle y, x \rangle \in R \}$

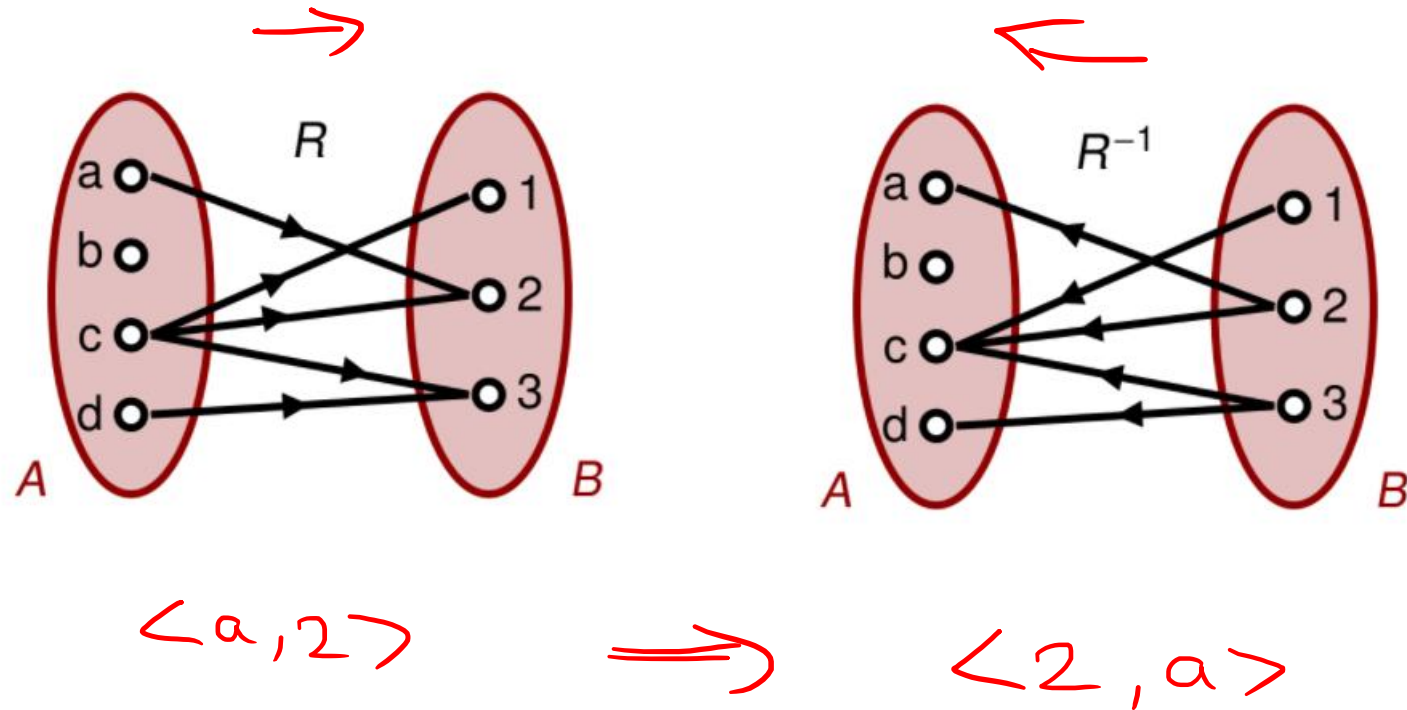
Example:

- R is the relation *HasFavoriteColor* Person x Color
- Then R^{-1} is the relation *IsFavoriteColorof* Color x Person

Inverse Relations

Definition:

The inverse of a relation R is $R^{-1} := \{ \langle x, y \rangle : \langle y, x \rangle \in R \}$



Composite Relations

Definition:

The composition of R and S: $R \circ S$ (R after S)

Is the relation $R \circ S = \{ \langle x, z \rangle : x S y \text{ and } y R z, \text{ for some } y \}$

Example

$R := \text{IsFriendOf} \circ \text{IsUncleOf}$

$\Rightarrow R = \{ \langle x, z \rangle : x \text{ is an uncle of a friend of } z \}$ (note the order!)

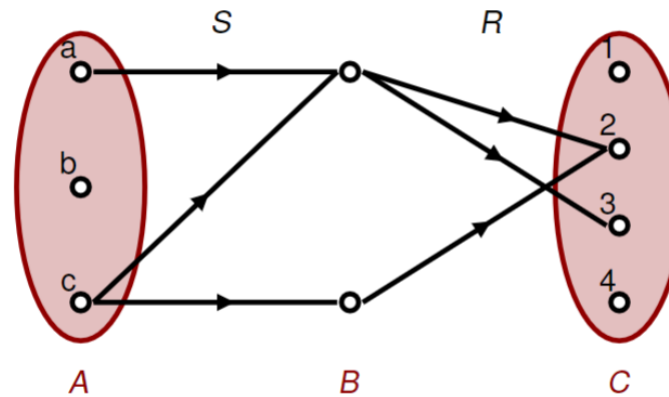
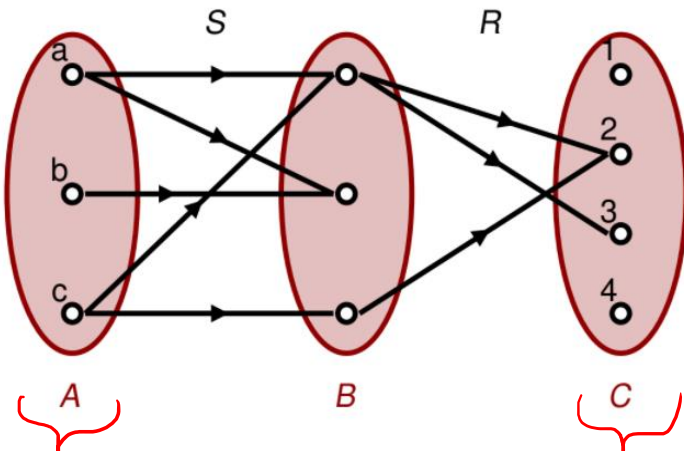


Composite Relations

Definition:

The composition of R and S : $R \circ S$ (R after S)

Is the relation $R \circ S = \{ \langle \underline{x}, \underline{z} \rangle : x S y \text{ and } y R z, \text{ for some } y \}$



$$R \circ S = \{ \langle a, 2 \rangle, \langle a, 3 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle \}$$

Composite Relations

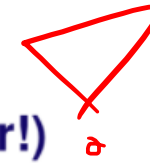
Definition:

The composition of R and S: $R \circ S$ (R after S)

Is the relation $R \circ S = \{ \langle x, z \rangle : x S y \text{ and } y R z, \text{ for some } y \}$

Composition is associative: $(R \circ S) \circ T = R \circ (S \circ T)$

The inverse of $R \circ S$ is $(R \circ S)^{-1} = \underbrace{S^{-1}} \circ \underbrace{R^{-1}}$ (note the order!)



Relation Properties

A relation can have a few properties:

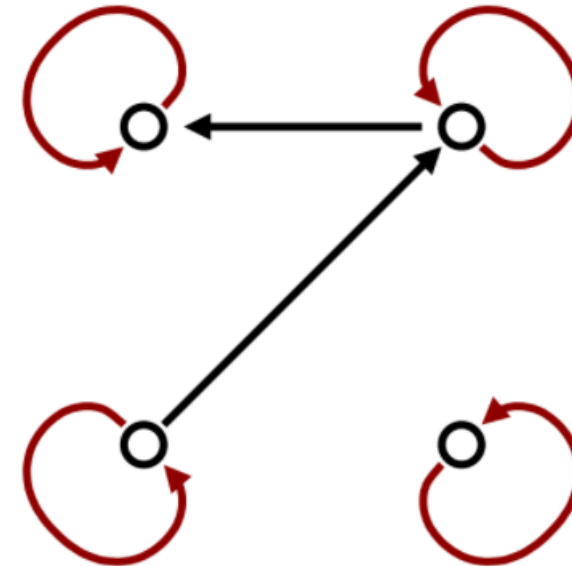
- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

Relation Properties

A relation can have a few properties:

- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

Reflexivity



for all elements x :
 $x R x$

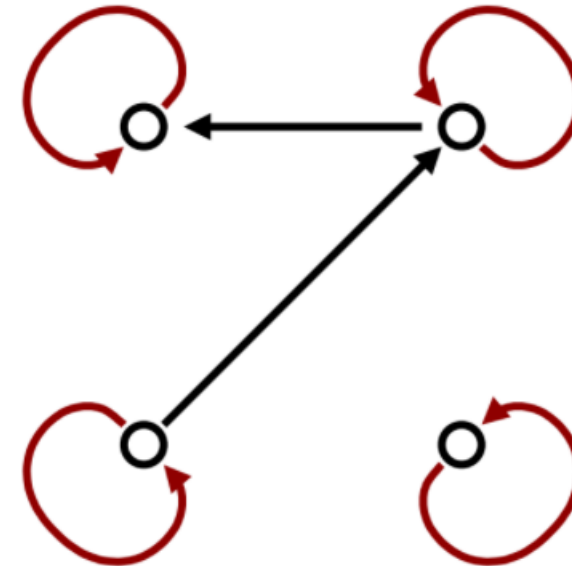
Relation Properties

A relation can have a few properties:

- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

$\models$ is *reflexive*: $\phi \models \phi$.

Reflexivity



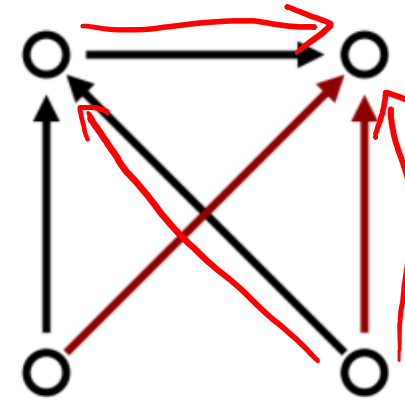
for all elements x :
 $x R x$

Relation Properties

A relation can have a few properties:

- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

Transitivity



for all elements x, y, z :
 $x R y \wedge y R z \rightarrow x R z$

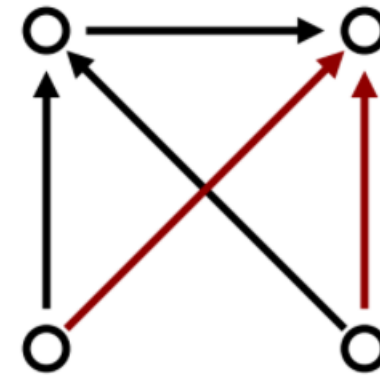
Relation Properties

A relation can have a few properties:

- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

$\models$ is *transitive*: If $\phi \models \psi$ and $\psi \models \chi$, then $\phi \models \chi$.

Transitivity



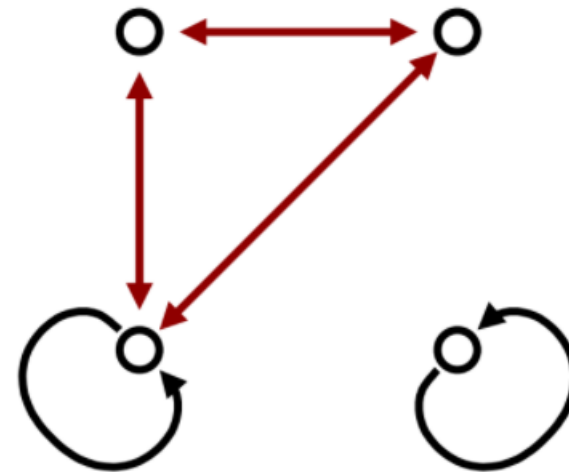
for all elements x, y, z :
 $x R y \wedge y R z \rightarrow x R z$

Relation Properties

A relation can have a few properties:

- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

Symmetry → double arrows!



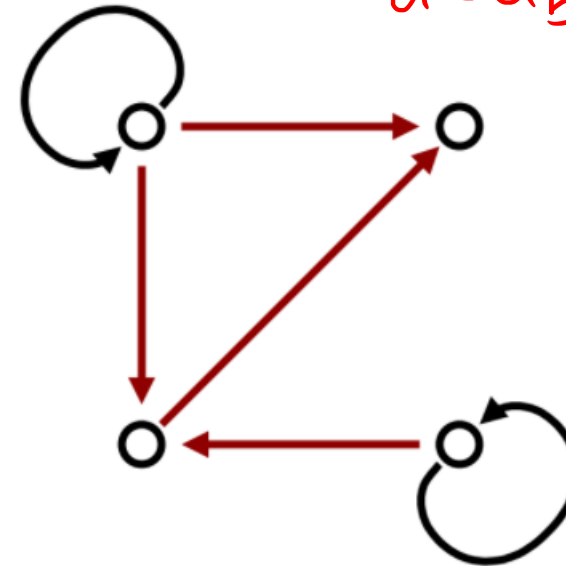
for all elements x, y :
 $x R y \rightarrow y R x$

Relation Properties

A relation can have a few properties:

- Reflexivity
- Transitivity
- Symmetry
- Anti-symmetry

Anti-symmetry → no double arrows!



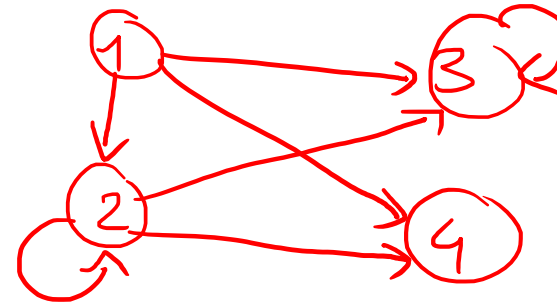
for all elements $x \neq y$:
 $x R y \rightarrow \neg y R x$

Relation Properties: Exercise!

6. Relations (6+5+6 points) In the set $V = \{1, 2, 3, 4\}$ consider the relations

$$R = \{\langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle, \langle 2, 4 \rangle, \langle 3, 3 \rangle\},$$

$$S = \{\langle 1, 1 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle\}.$$



~~a) Write down the matrix representation of the relation R .~~

b) Is the relation R transitive? Is it reflexive? (Explain your answers.)

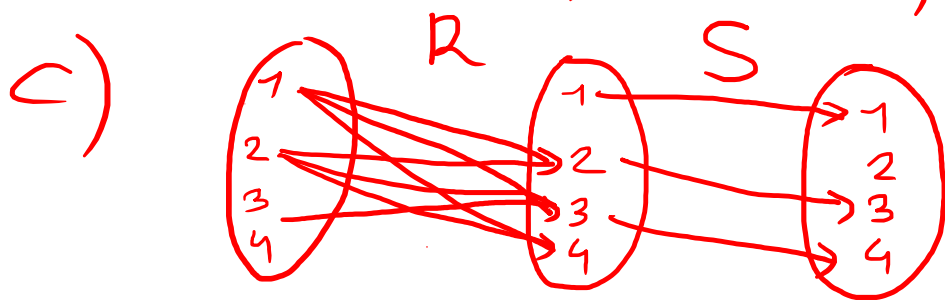
c) Draw a Venn diagram depicting the relation $S \circ R$.

b) not reflexive: $\langle 1, 1 \rangle \notin R$ OR $\langle 4, 4 \rangle \notin R$

is transitive: $\langle 1, 2 \rangle \in R \wedge \langle 2, 3 \rangle \in R \Rightarrow \langle 1, 3 \rangle \in R$

and $\langle 1, 2 \rangle \in R \wedge \langle 2, 4 \rangle \in R \Rightarrow \langle 1, 4 \rangle \in R$

(also antisymmetric)



$$S \circ R = \{\langle 1, 3 \rangle, \langle 2, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 4 \rangle, \langle 3, 4 \rangle\}$$



Relation Properties: Exercise!

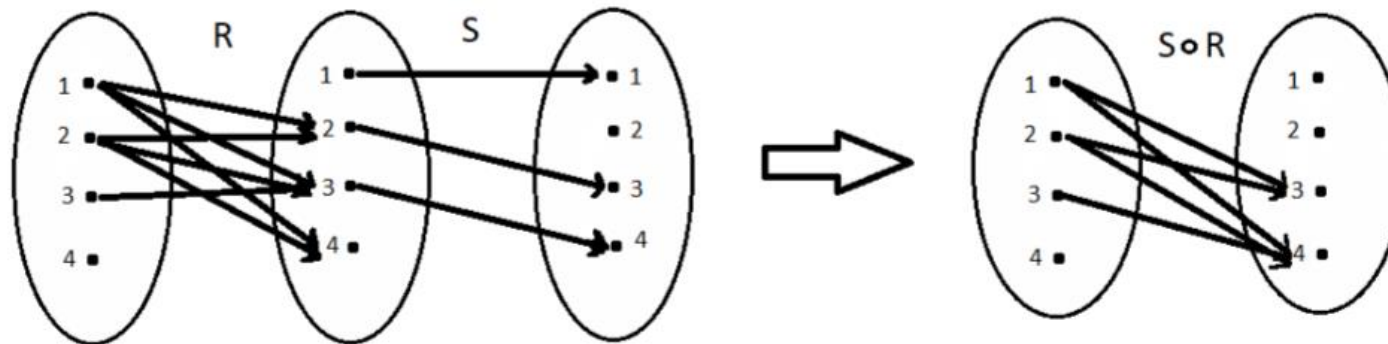
6. Relations (6+5+6 points) In the set $V = \{1, 2, 3, 4\}$ consider the relations

$$R = \{\langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle, \langle 2, 4 \rangle, \langle 3, 3 \rangle\},$$

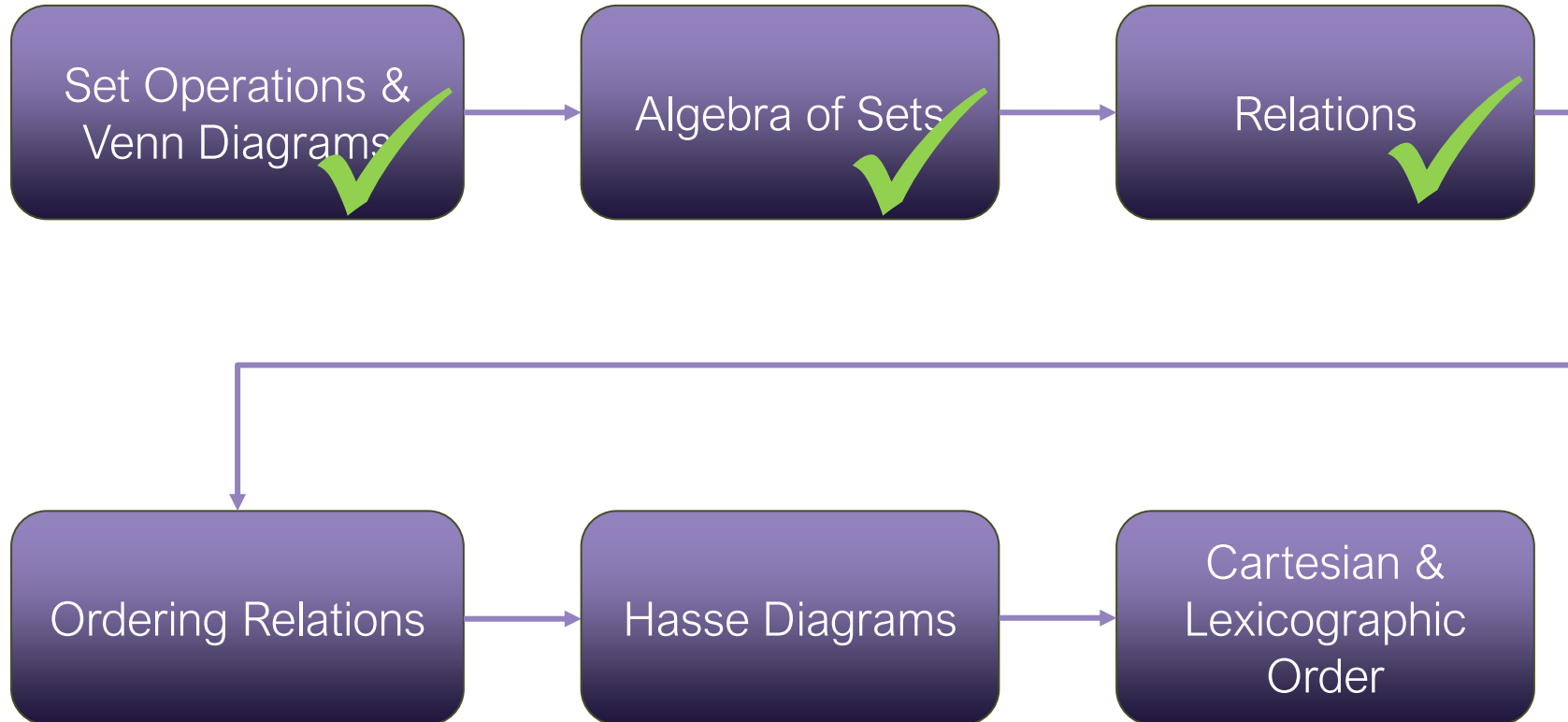
$$S = \{\langle 1, 1 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle\}.$$

- ~~a) Write down the matrix representation of the relation R .~~
- b) Is the relation R transitive? Is it reflexive? (Explain your answers.)
- c) Draw a Venn diagram depicting the relation $S \circ R$.

- b) – R is *not* reflexive, since $\langle 1, 1 \rangle \notin R$ or $\langle 4, 4 \rangle \notin R$
- R is transitive, since $\langle 1, 2 \rangle \in R \wedge \langle 2, 3 \rangle \in R \Rightarrow \langle 1, 3 \rangle \in R$ and $\langle 1, 2 \rangle \in R \wedge \langle 2, 4 \rangle \in R \Rightarrow \langle 1, 4 \rangle \in R$.
- c) We find $S \circ R = \{\langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 3 \rangle, \langle 2, 4 \rangle, \langle 3, 4 \rangle\}$



Planning



Ordering Relations

Ordering Relations

A partial order on V is a relation of type $V \times V$ that is reflexive and transitive and anti-symmetric

Reminder



▶ reflexivity

$$x R x$$



▶ transitivity

$$x R y \wedge y R z \rightarrow x R z$$



▶ anti-symmetry

$$(x \neq y \wedge x R y) \rightarrow \neg y R x$$

$$x R y \wedge y R x \rightarrow x = y$$



Ordering Relations

A **partial order** on V is a relation of type $V \times V$ that is reflexive and transitive and anti-symmetric

Consider a partial order R on a set V . Then

- ▶ $x, y \in V$ are **comparable** if $x R y$ or $y R x$
- ▶ R is a **linear** (or **total**) **order** if all $x, y \in V$ are comparable

partial: $\subseteq$

$\mathcal{P}(\{1, 2\})$ $\{1\} \subseteq \{1, 2\}$

$\{1\} \not\subseteq \{2\}$

$\{2\} \not\subseteq \{1\}$

total:

$R: \leq$ every int $x, y: y \leq x$ OR $x \leq y \Rightarrow$ total

lexicographic:

every letter

$x, y: x$ before y OR y before x

Ordering Relations

A **partial order** on V is a relation of type $V \times V$ that is reflexive and transitive and anti-symmetric

partial:
 $\subseteq$
 $A \subseteq A$

Consider a partial order R on a set V . Then

- ▶ $x, y \in V$ are **comparable** if $x R y$ or $y R x$
- ▶ R is a **linear** (or **total**) **order** if all $x, y \in V$ are comparable
- ▶ the **strict partial order** S corresponding to R is defined by

$$x S y \iff x R y \text{ and } x \neq y$$

A strict partial order is irreflexive and transitive and anti-symmetric
 $\forall x \neg (x S x)$

strict partial:
 $<$
 $x < x$
 not possible

Ordering Relations: Exercise

Exam Question: Final 2022

6. Relations (3+5+4 points)

In the set $V = \{1, 2, 3, 4\}$ consider the relation R given by the set of pairs

$\{\langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle\}$.

b) Show that R is an ordering relation which is not total.

b) Reflexive: $\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle$

transitive: $\langle 1, 2 \rangle \wedge \langle 2, 3 \rangle \rightarrow \langle 1, 3 \rangle$

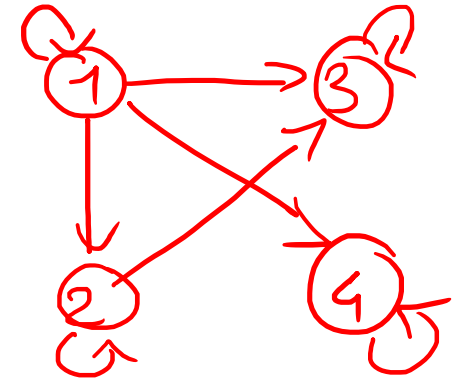
anti-symmetric: $(x, y) \notin R$ or $(y, x) \notin R$ whenever $x \neq y$
 $\Rightarrow$ partial order

not total: $\langle 3, 4 \rangle, \langle 4, 3 \rangle \notin R$ so not all x, y are comparable

A partial order on V is a relation of type $V \times V$ that is reflexive and transitive and anti-symmetric

Consider a partial order R on a set V . Then

- ▶ $x, y \in V$ are **comparable** if $x R y$ or $y R x$
- ▶ R is a **linear** (or **total**) **order** if all $x, y \in V$ are comparable



Ordering Relations: Exercise

Exam Question: Final 2022

6. Relations (3+5+4 points)

In the set $V = \{1, 2, 3, 4\}$ consider the relation R given by the set of pairs

$$\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (3, 3), (4, 4)\}.$$

b) Show that R is an ordering relation which is not total.

R is a partial order since:

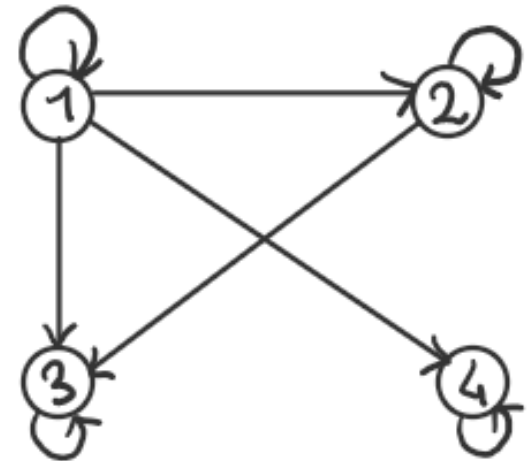
- R is reflexive: $\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle$
- R is transitive: $\langle 1, 2 \rangle \wedge \langle 2, 3 \rangle \rightarrow \langle 1, 3 \rangle$
- R is anti-symmetric: $(x, y) \notin R$ or $(y, x) \notin R$ whenever $x \neq y$

R is **not** a total order since: $\langle 3, 4 \rangle, \langle 4, 3 \rangle \notin R$, so not all $x, y \in V$ are comparable

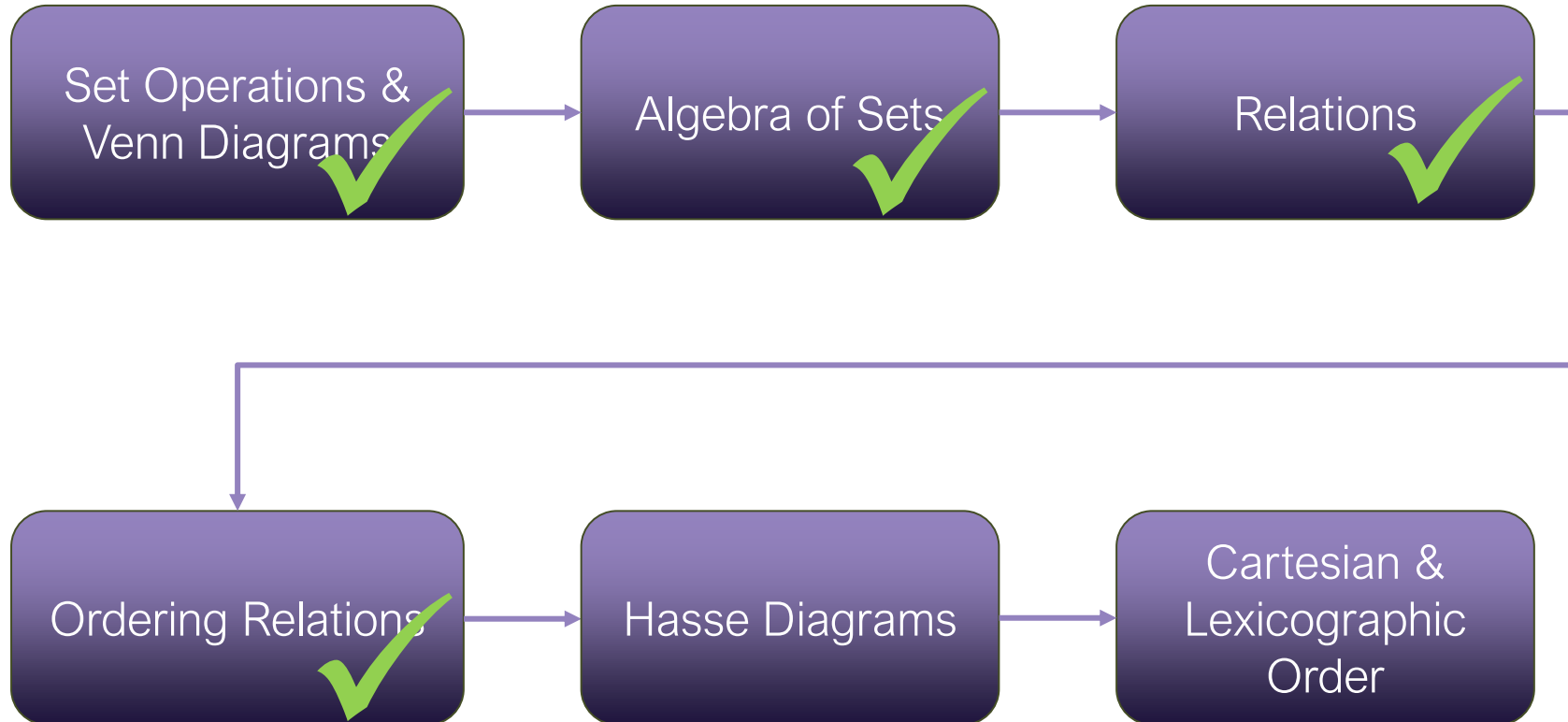
A **partial order** on V is a relation of type $V \times V$ that is reflexive and transitive and anti-symmetric

Consider a partial order R on a set V . Then

- $x, y \in V$ are **comparable** if $x R y$ or $y R x$
- R is a **linear** (or **total**) **order** if all $x, y \in V$ are comparable

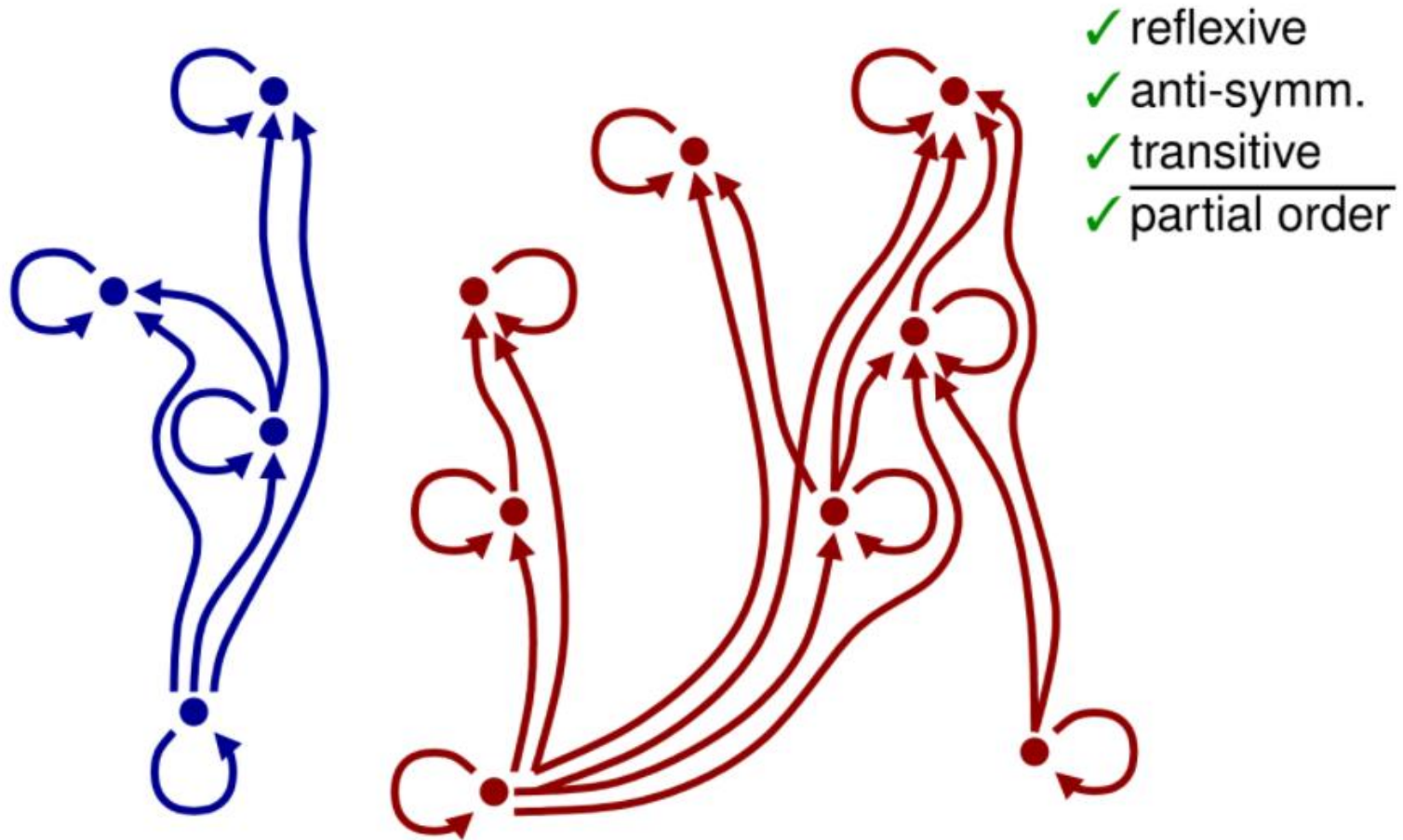


Planning

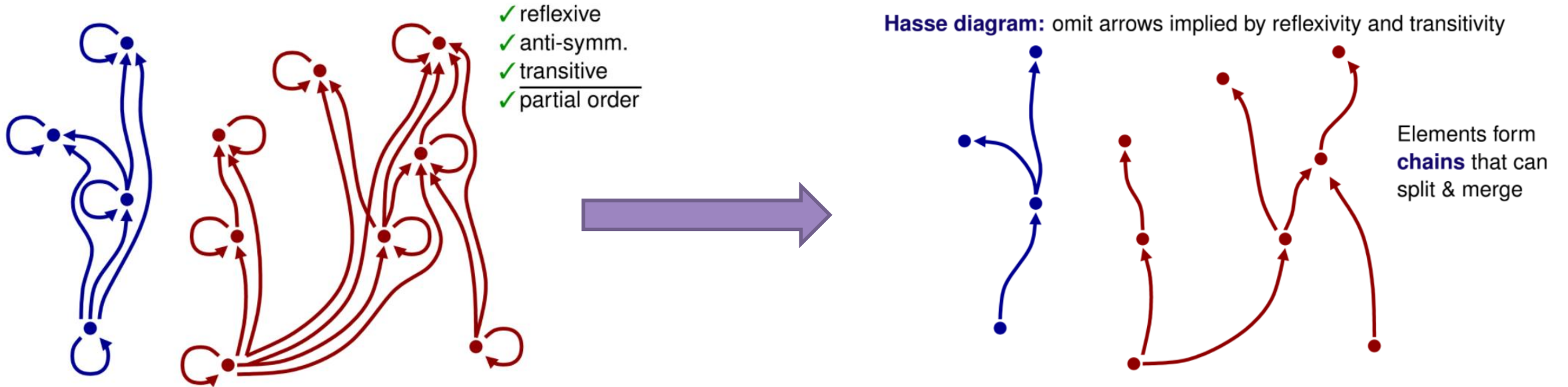


Hasse Diagrams

Ordering Relations



Hasse Diagram



Hasse Diagram

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$)

Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$

$R := \text{IsDivisorOf}$

$$G_1 = \{2, 3, 4, 6, 12\}$$

$$G_2 = \{4, 6, 12\}$$

$$G_3 = \{6, 12\}$$

$$G_4 = \{12\}$$

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

$1 \neq 1$

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
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Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$

$R := \text{IsDivisorOf}$

$$G_1 = \{2, 3, 4, 6, 12\}$$

$$G_2 = \{4, 6, 12\}$$

$$G_3 = \{6, 12\}$$

$$G_4 = \{12\}$$

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

H_1

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y: y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$ $R := \text{IsDivisorOf}$

$$\begin{aligned} G_1 &= \{2, 3, 4, 6, 12\} & H_1 &= \{2, 3\} \\ G_2 &= \{4, 6, 12\} \\ G_3 &= \{6, 12\} \\ G_4 &= \{12\} \\ G_6 &= \{12\} \\ G_{12} &= \{\} \end{aligned}$$

remove $\bigcup_{y \in G_1} G_y$ from G_1

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$

$R := \text{IsDivisorOf}$

4, 6

$$G_1 = \{2, 3, 4, 6, 12\}$$

$$G_2 = \{4, 6, 12\}$$

$$G_3 = \{6, 12\}$$

$$G_4 = \{12\}$$

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

$$H_1 = \{2, 3\}$$

$$H_2$$

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$ $R := \text{IsDivisorOf}$

$$G_1 = \{2, 3, 4, 6, 12\}$$

$$H_1 = \{2, 3\}$$

$$G_2 = \{4, 6, 12\}$$

$$H_2 = \{4, 6\}$$

$$G_3 = \{6, 12\}$$

$$G_4 = \{12\}$$

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

remove $\bigcup_{y \in G_2} G_y$ from G_2

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$ $R := \text{IsDivisorOf}$

$$G_1 = \{2, 3, 4, 6, 12\}$$

$$G_2 = \{4, 6, 12\}$$

$$G_3 = \{6, 12\}$$

$$G_4 = \{12\}$$

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

$$H_1 = \{2, 3\}$$

$$H_2 = \{4, 6\}$$

$$H_3 = \{6\}$$

$$H_4 = \{12\}$$

$$H_6 = \{12\}$$

$$H_{12} = \{\}$$

Algorithm (Hasse diagram for a given partial order R on V)

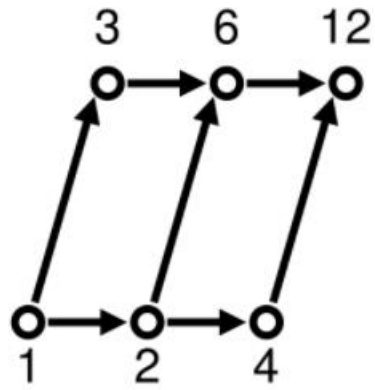
1. for all $x \in V$ $G_x := \{y: y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Hasse Diagram: Example

Example $V := \{1, 2, 3, 4, 6, 12\}$

$R := \text{IsDivisorOf}$



$$H_1 = \{2, 3\}$$

$$H_2 = \{4, 6\}$$

$$H_3 = \{6\}$$

$$H_4 = \{12\}$$

$$H_6 = \{12\}$$

$$H_{12} = \{\}$$

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y: y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Hasse Diagram: Exercise!

Exam Question: 2020

4. Ordering relations (8 + 2 + 2 points)

Consider the ordering relation *IsDivisorOf* on the set *A* defined by

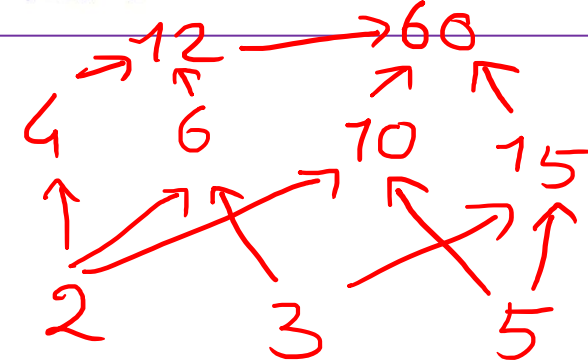
$$A := \{2, 3, 4, 5, 6, 10, 12, 15, 60\}.$$

- (a) Use the algorithm that you have learned to construct the Hasse diagram that represents the *IsDivisorOf* ordering relation on *A*: explicitly write down the sets G_x and H_x (with $x \in A$) obtained in the construction, and then draw the Hasse diagram.

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

($\bigcup_{y \in G_x} G_y$ is the **union** of all sets G_y with $y \in G_x$)



$$\begin{aligned} G_2 &= \{4, 6, 10, 12, 60\} \\ G_3 &= \{6, 15, 60\} \\ G_4 &= \{12, 60\} \\ G_5 &= \{10, 15, 60\} \\ G_6 &= \{12, 60\} \\ G_{10} &= \{60\} \\ G_{12} &= \{60\} \\ G_{15} &= \{60\} \end{aligned}$$

$$G_{60} = \emptyset$$

$$\begin{aligned} H_2 &= \{4, 6, 10\} \\ H_3 &= \{6, 15\} \\ H_4 &= \{12\} \\ H_5 &= \{10, 15\} \\ H_6 &= \{12\} \\ H_{10} &= \{60\} \end{aligned}$$

$$\begin{aligned} H_{12} &= \{60\} \\ H_{15} &= \{60\} \\ H_{60} &= \emptyset \end{aligned}$$

Hasse Diagram: Exercise!

Exam Question: 2020

4. Ordering relations (8 + 2 + 2 points)

Consider the ordering relation *IsDivisorOf* on the set A defined by

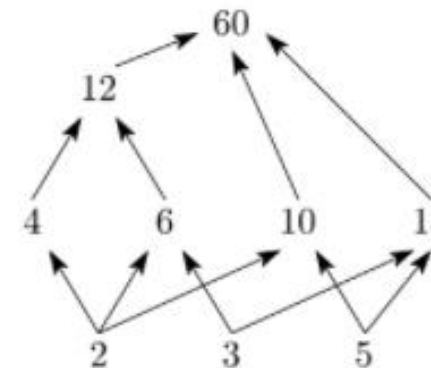
$$A := \{2, 3, 4, 5, 6, 10, 12, 15, 60\}.$$

- (a) Use the algorithm that you have learned to construct the Hasse diagram that represents the *IsDivisorOf* ordering relation on A : explicitly write down the sets G_x and H_x (with $x \in A$) obtained in the construction, and then draw the Hasse diagram.

Solution: The sets G_x and H_x are as follows:

$G_2 = \{4, 6, 10, 12, 60\}$	$H_2 = \{4, 6, 10\}$
$G_3 = \{6, 12, 15, 60\}$	$H_3 = \{6, 15\}$
$G_4 = \{12, 60\}$	$H_4 = \{12\}$
$G_5 = \{10, 15, 60\}$	$H_5 = \{10, 15\}$
$G_6 = \{12, 60\}$	$H_6 = \{12\}$
$G_{10} = \{60\}$	$H_{10} = \{60\}$
$G_{12} = \{60\}$	$H_{12} = \{60\}$
$G_{15} = \{60\}$	$H_{15} = \{60\}$
$G_{60} = \emptyset$	$H_{60} = \emptyset$

so this is the corresponding Hasse diagram:

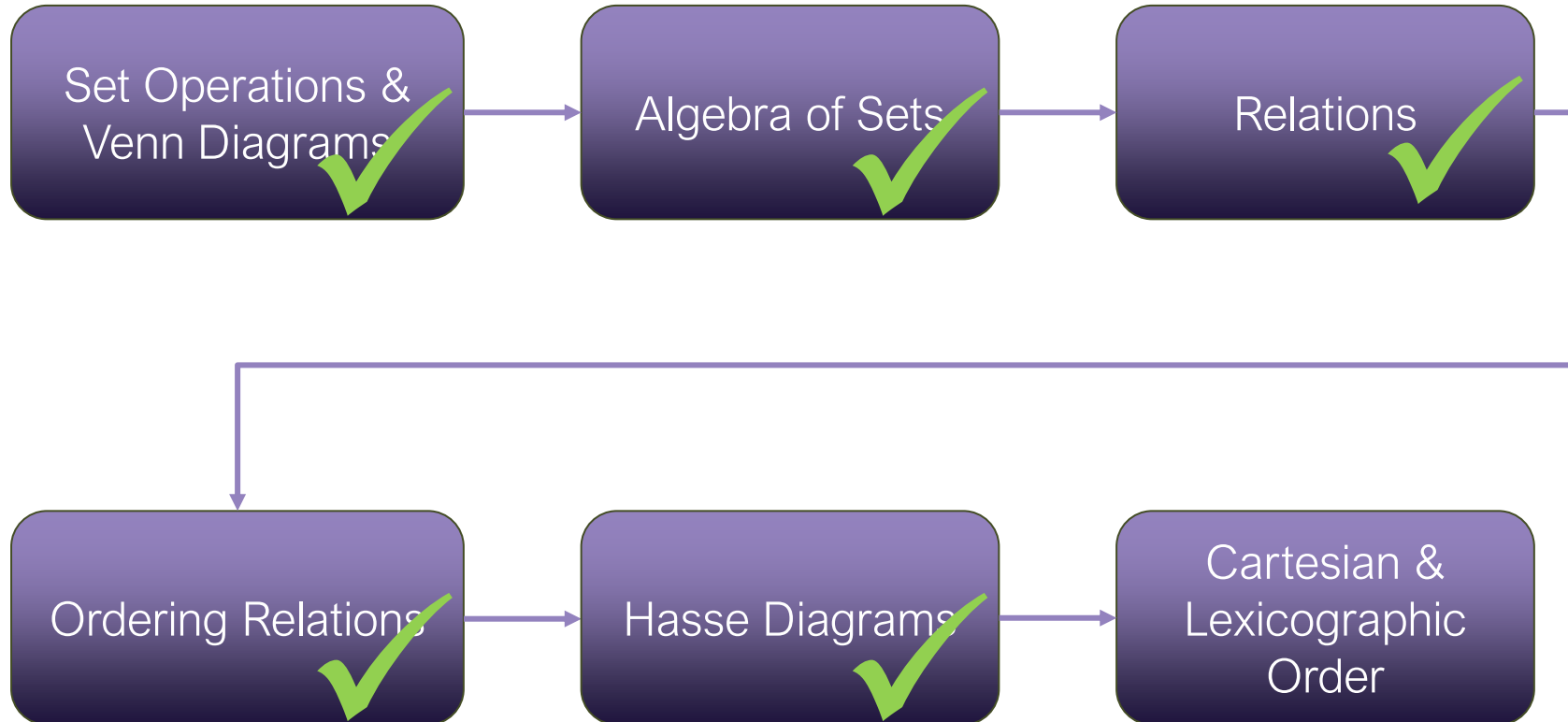


Algorithm (Hasse diagram for a given partial order R on V)

- for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
- for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
- for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$

Planning



Cartesian & Lexicographic Order

Evaluation form



Step 1

Scan the QR code
or check your email



Step 2

Answer the 5
questions



Step 3

Sit back & relax



Cartesian and Lexicographic Order

Suppose we have partial orders $\leq_A$ on A and $\leq_B$ on B

Lexicographic order (A takes priority over B)

$$\langle a_1, b_1 \rangle \leq \langle a_2, b_2 \rangle \iff (a_1 <_A a_2) \vee (a_1 = a_2 \wedge b_1 \leq_B b_2)$$

- $ab < ac$
 - $ab < bb$
 - $ac < bb$
- } $ab < ac < bb$

Cartesian and Lexicographic Order

Suppose we have partial orders $\leq_A$ on A and $\leq_B$ on B

Lexicographic order (A takes priority over B)

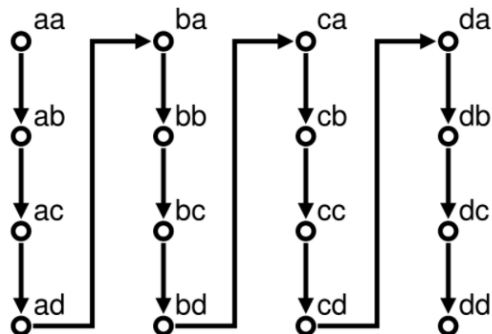
$$\langle a_1, b_1 \rangle \leq \langle a_2, b_2 \rangle \iff (a_1 <_A a_2) \vee (a_1 = a_2 \wedge b_1 \leq_B b_2)$$

Lexicographic order = dictionary order

Example

Take $A := \{a, b, c, d\}$ with $\leq_A$ the usual alphabetic ordering of letters

Lexicographic order on $A \times A$:



Cartesian and Lexicographic Order

Suppose we have partial orders $\leq_A$ on A and $\leq_B$ on B

Lexicographic order (A takes priority over B) 

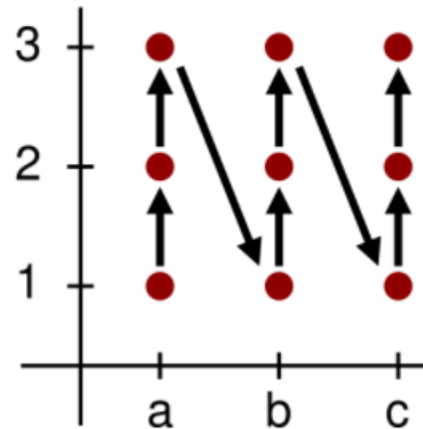
$$\langle a_1, b_1 \rangle \leq \langle a_2, b_2 \rangle \iff \underbrace{(a_1 <_A a_2)}_{\text{red arrow}} \vee \underbrace{(a_1 = a_2 \wedge b_1 \leq_B b_2)}_{\text{red underline}}$$

Example

$$A = \{a, b, c\}$$

$$B = \{1, 2, 3\}$$

with the usual order

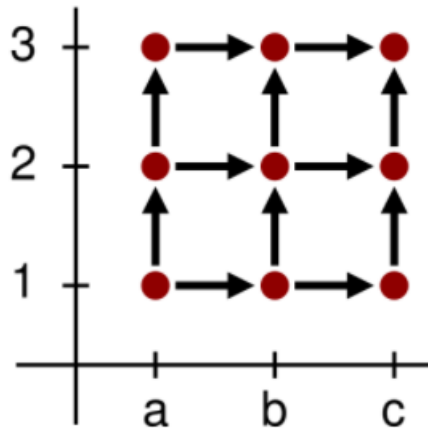


Cartesian and Lexicographic Order

Suppose we have partial orders $\leq_A$ on A and $\leq_B$ on B

Cartesian order (A and B treated equally) 

$$\langle a_1, b_1 \rangle \leq \langle a_2, b_2 \rangle \iff a_1 \leq_A a_2 \text{ and } b_1 \leq_B b_2$$



$$A = \{a, b, c\}$$

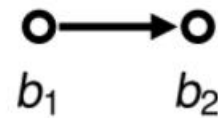
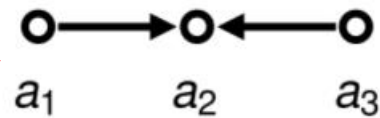
$$B = \{1, 2, 3\}$$

$$\langle a, 1 \rangle$$

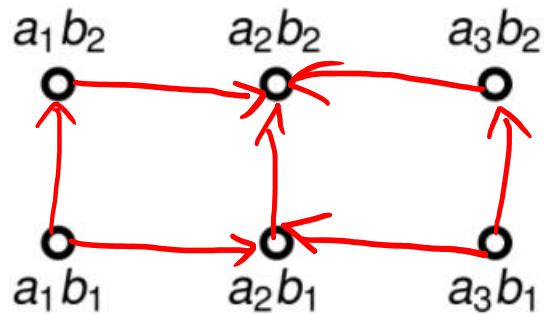
$$\searrow \langle b, 1 \rangle \quad \searrow \langle a, 2 \rangle$$

Cartesian and Lexicographic Order

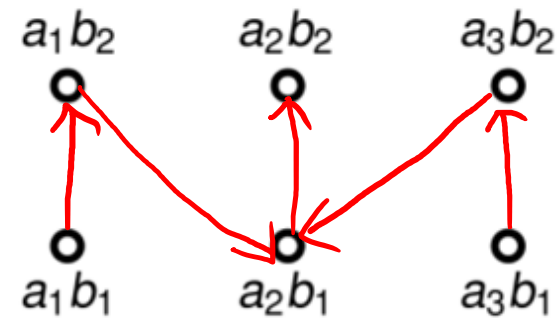
Let $A := \{a_1, a_2, a_3\}$ and $B := \{b_1, b_2\}$ with partial orders $\leq_A$ and $\leq_B$ on A and B described by the following Hasse diagrams:



Draw Hasse diagrams for the Cartesian and lexicographic orders on $A \times B$ (induced by $\leq_A$ and $\leq_B$) by adding arrows to this picture:



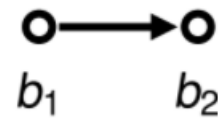
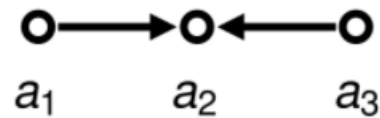
CARTESIAN



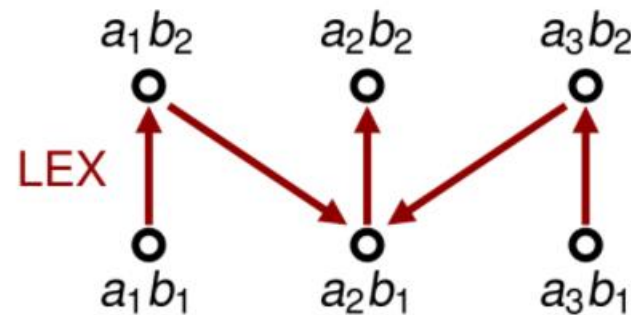
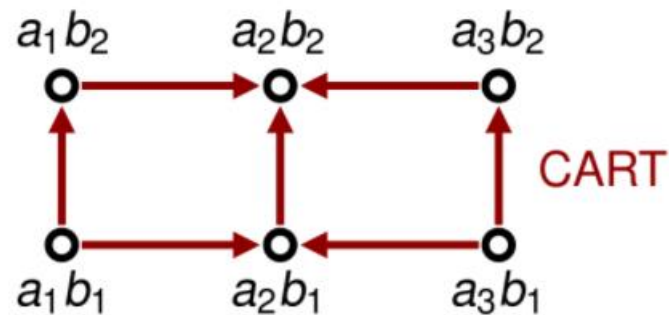
LEX

Cartesian and Lexicographic Order

Let $A := \{a_1, a_2, a_3\}$ and $B := \{b_1, b_2\}$ with partial orders $\leq_A$ and $\leq_B$ on A and B described by the following Hasse diagrams:



Draw Hasse diagrams for the Cartesian and lexicographic orders on $A \times B$ (induced by $\leq_A$ and $\leq_B$) by adding arrows to this picture:

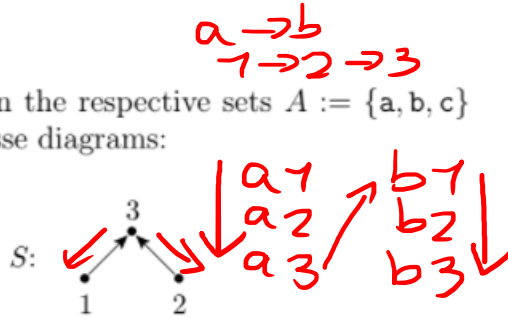
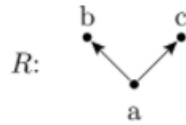


Cartesian and Lexicographic Order: Exercise!

Exam Question: resit 2019

7. Ordering relations (8 + 3 points)

Let R and S be the partial ordering relations on the respective sets $A := \{a, b, c\}$ and $B := \{1, 2, 3\}$, specified by the following Hasse diagrams:



Consider the product set

$$A \times B = \{\langle a, 1 \rangle, \langle a, 2 \rangle, \langle a, 3 \rangle, \langle b, 1 \rangle, \langle b, 2 \rangle, \langle b, 3 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle\}.$$

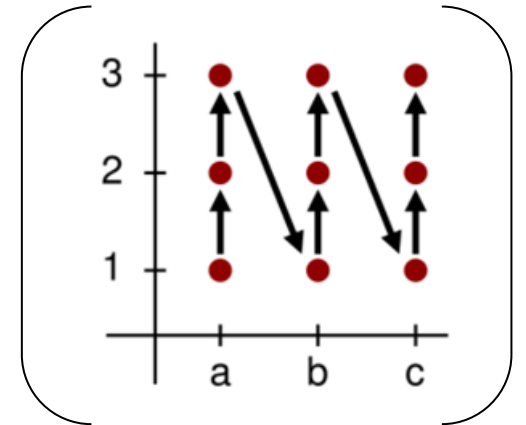
We are interested in the lexicographic order on $A \times B$ induced by the partial orders R on A , and S on B .

- (a) Use the algorithm you have learned to construct the Hasse diagram of the lexicographic order on the set $A \times B$: write down the sets G_x and H_x obtained in the construction, and use them to draw the Hasse diagram. (If you prefer, you can simplify the notation by writing a_1 for $\langle a, 1 \rangle$, a_2 for $\langle a, 2 \rangle$, and so on.)

Algorithm (Hasse diagram for a given partial order R on V)

- for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
- for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
- for all $x \in V$ draw an arrow from x to every $y \in H_x$

$\left(\bigcup_{y \in G_x} G_y \right)$ is the **union** of all sets G_y with $y \in G_x$



Handwritten red notes for H_x sets:

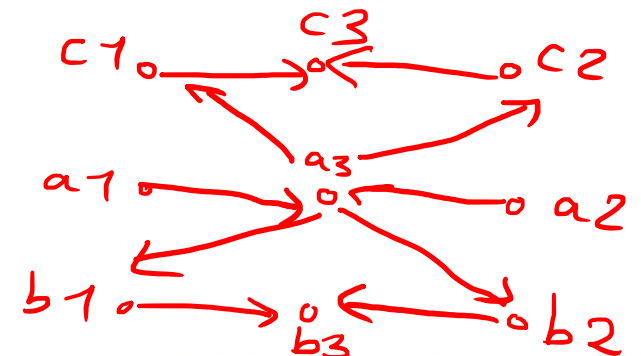
- $H_{a_1} = \{a_3\}$
- $H_{a_2} = \{a_3\}$
- $H_{a_3} = \{b_1, b_2, c_1, c_2\}$
- $H_{b_1} = \{b_3\}$
- $H_{b_2} = \{b_3\}$
- $H_{b_3} = \{c_1, c_2\}$
- $H_{c_1} = \{c_3\}$
- $H_{c_2} = \{c_3\}$
- $H_{c_3} = \{\}$

Handwritten red notes for G_x sets:

- $G_{a_1} = \{a_3, b_1, b_2, b_3, c_1, c_2, c_3\}$
- $G_{a_2} = \{a_3, b_1, b_2, b_3, c_1, c_2, c_3\}$
- $G_{a_3} = \{b_1, b_2, b_3, c_1, c_2, c_3\}$
- $G_{b_1} = \{b_3\}$

Handwritten red notes for G_x sets (continued):

- $G_{b_2} = \{b_3\}$
- $G_{b_3} = \{c_1, c_2\}$
- $G_{c_1} = \{c_3\}$
- $G_{c_2} = \{c_3\}$
- $G_{c_3} = \{\}$



Cartesian and Lexicographic Order: Exercise!

Exam Question: resit 2019

7. Ordering relations (8 + 3 points)

Let R and S be the partial ordering relations on the respective sets $A := \{a, b, c\}$ and $B := \{1, 2, 3\}$, specified by the following Hasse diagrams:



Consider the product set

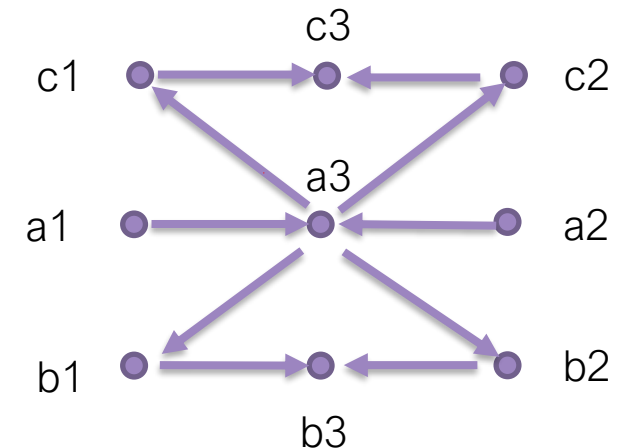
$$A \times B = \{\langle a, 1 \rangle, \langle a, 2 \rangle, \langle a, 3 \rangle, \langle b, 1 \rangle, \langle b, 2 \rangle, \langle b, 3 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle\}.$$

We are interested in the lexicographic order on $A \times B$ induced by the partial orders R on A , and S on B .

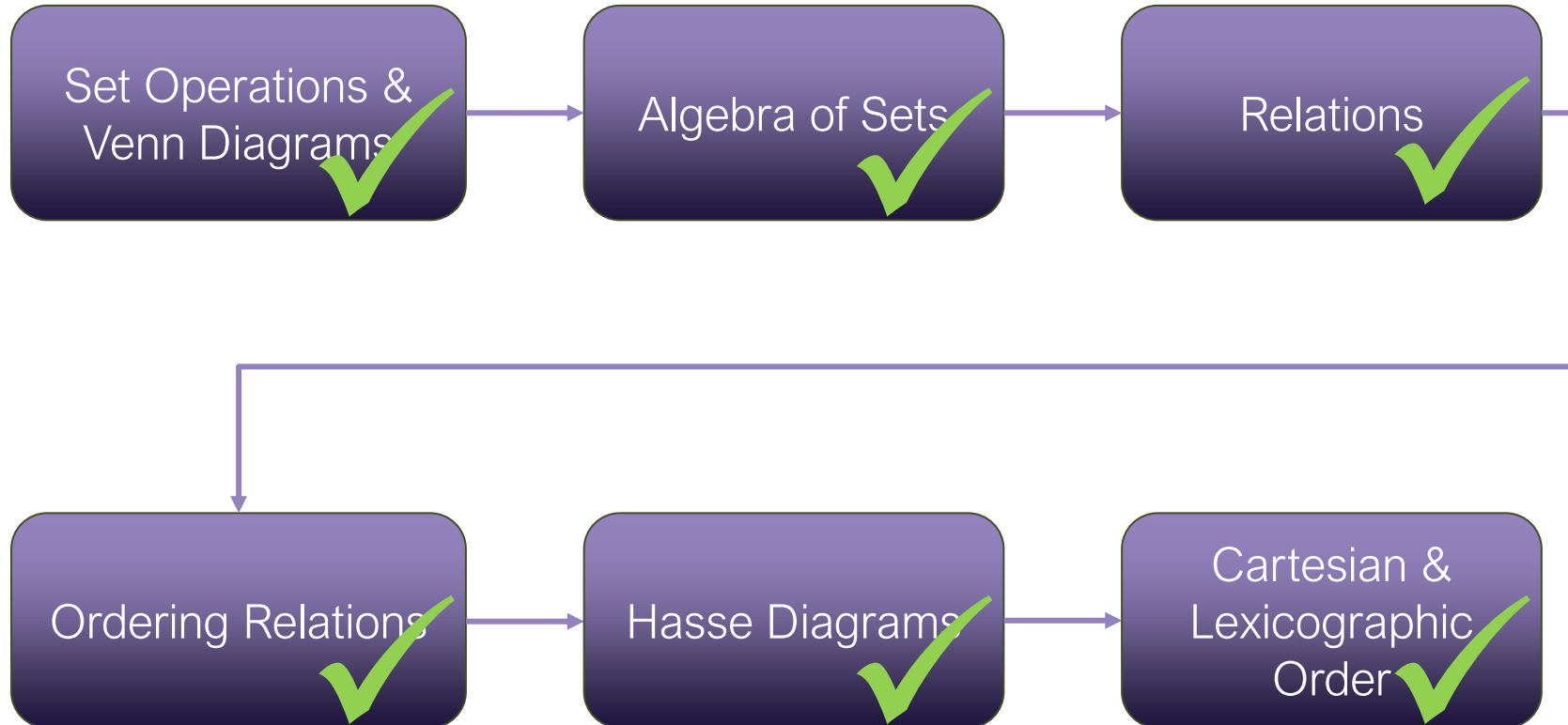
- (a) Use the algorithm you have learned to construct the Hasse diagram of the lexicographic order on the set $A \times B$: write down the sets G_x and H_x obtained in the construction, and use them to draw the Hasse diagram. (If you prefer, you can simplify the notation by writing $a1$ for $\langle a, 1 \rangle$, $a2$ for $\langle a, 2 \rangle$, and so on.)

Answer:

$G_{a1} = \{a3, b1, b2, b3, c1, c2, c3\}$	$H_{a1} = \{a3\}$
$G_{a2} = \{a3, b1, b2, b3, c1, c2, c3\}$	$H_{a2} = \{a3\}$
$G_{a3} = \{b1, b2, b3, c1, c2, c3\}$	$H_{a3} = \{b1, b2, c1, c2, c3\}$
$G_{b1} = \{b3\}$	$H_{b1} = \{b3\}$
$G_{b2} = \{b3\}$	$H_{b2} = \{b3\}$
$G_{b3} = \{\}$	$H_{b3} = \{\}$
$G_{c1} = \{c3\}$	$H_{c1} = \{c3\}$
$G_{c2} = \{c3\}$	$H_{c2} = \{c3\}$
$G_{c3} = \{\}$	$H_{c3} = \{\}$



Finished!

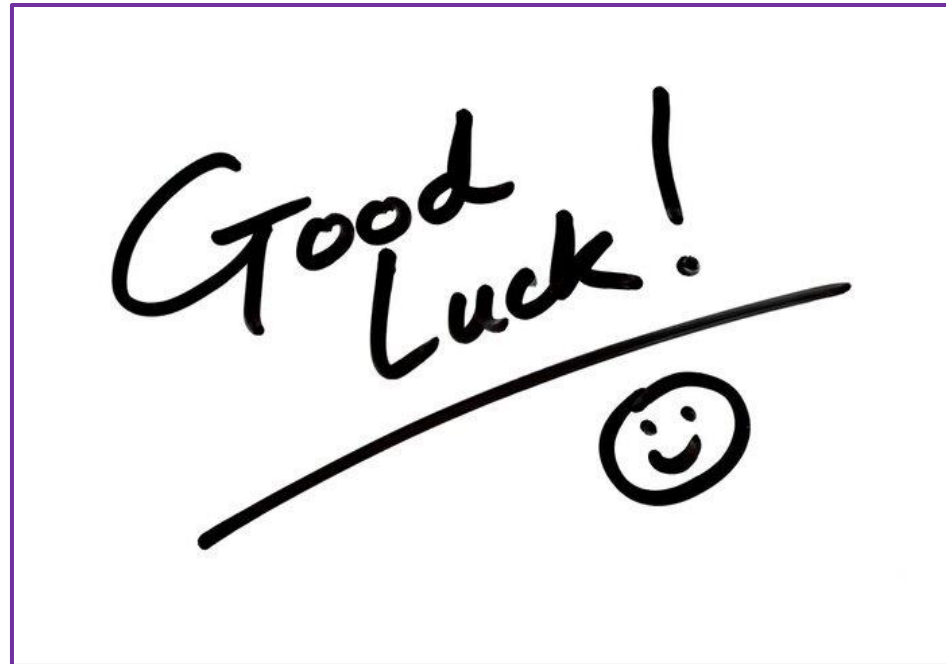


Questions?

Join the WhatsApp group for slides! Invite link should be in your mail

If you have any questions, ask them in the WhatsApp group :)

Thank you for your attention and participation and good luck with your exam!



Evaluation form



Step 1

Scan the QR code
or check your email



Step 2

Answer the 5
questions



Step 3

Sit back & relax

